
Normalization Layers as Implicit Spectral Filters: Optimizer Regime Shapes Frequency Bias Across Vision Architectures

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Normalization layers appear in virtually every vision architecture, yet how they
2 reshape the frequency spectrum of learned representations, and what factors gov-
3 ern the direction of this reshaping, remains poorly understood. We present a
4 cross-architecture, cross-optimizer diagnostic measuring the spectral gap $\Delta_{H/L}$
5 (the median difference in high-to-low frequency energy ratio between Batch Nor-
6 malization (BN) and Layer Normalization (LN) variants) across six architectures
7 from four families at $\geq 64 \times 64$ resolution, under both SGD and AdamW training.
8 Our central observation is that optimizer regime is a directionally consistent corre-
9 late of spectral bias direction: all five AdamW-trained configurations at $\geq 64 \times 64$
10 resolution yield non-positive $\Delta_{H/L}$, while SGD training on CNN-family architec-
11 tures produces positive $\Delta_{H/L}$. A controlled LINCHPIN experiment (same ResNet-
12 50, identical data, only the optimizer changed) shifts $\Delta_{H/L}$ from +0.454 under
13 SGD to near-zero under AdamW; a learning-rate gradient control shows mono-
14 tonic decay of $\Delta_{H/L}$ with effective update magnitude. Extending SGD to non-
15 CNN architectures produces catastrophic block collapse rather than measurable
16 spectral gaps, further delimiting the conditions under which normalization-driven
17 spectral bias emerges. Taken together, these results suggest that normalization’s
18 spectral footprint is shaped primarily by effective update magnitude within the
19 optimizer–architecture interaction, rather than by normalization type alone.

20 1 Introduction

21 Batch Normalization (BN) Ioffe and Szegedy [2015] and Layer Normalization (LN) Ba et al. [2016]
22 are ubiquitous in modern vision architectures, yet their influence extends beyond training stability
23 into a less-explored domain: the frequency spectrum of learned representations. The spectral bal-
24 ance between low-frequency (smooth, global) and high-frequency (textured, local) energy has di-
25 rect practical consequences: high-frequency sensitivity correlates with texture bias and adversarial
26 vulnerability Yin et al. [2019], Geirhos et al. [2019]. Given that BN and LN operate on fundamen-
27 tally different statistical axes (batch-wise vs. channel-wise standardization), a natural expectation
28 is that the normalization–architecture interaction determines how spectral energy is redistributed:
29 BN should reshape frequencies differently in CNNs than in transformers, and the direction of this
30 reshaping should depend on the architectural inductive biases through which normalization oper-
31 ates Santurkar et al. [2018], Lubana et al. [2021].

32 This expectation is wrong, or at best incomplete. We present a systematic cross-architecture diag-
33 nostic spanning six architectures from four design families (ResNet-50 He et al. [2016], ConvNeXt-
34 T Liu et al. [2022], EfficientNet-B0 Tan and Le [2019] (CNNs), DeiT-S Dosovitskiy et al. [2021],
35 Touvron et al. [2021] (vision transformer), VMamba-S Gu and Dao [2024], Liu et al. [2024] (state-

space model), and MambaVision-T Hatamizadeh and Kautz [2025] (SSM–attention hybrid)), each trained with both BN and LN under SGD and AdamW. The central finding is that *optimizer regime, specifically effective update magnitude, is the primary determinant of the BN–LN spectral gap direction*. Under AdamW, all five configurations at $\geq 64 \times 64$ resolution yield non-positive $\Delta_{H/L}$ (sign test $p = 0.031$); the near-zero ResNet-50+AdamW anchor used in the LINCHPIN analysis is excluded from this count. This directional consistency holds regardless of whether the architecture is convolutional, attention-based, or state-space. Under SGD, CNN-family architectures produce strongly positive $\Delta_{H/L}$, meaning BN shifts representations toward higher frequencies relative to LN. Architecture family modulates magnitude but does not determine sign. We note that optimizer-induced training collapse precluded SGD configurations for non-CNN architectures, which limits cross-optimizer coverage outside the CNN family (§4.1).

The sharpest evidence comes from a controlled LINCHPIN experiment. On the same ResNet-50 with identical data and training duration, switching from SGD to AdamW collapses $\Delta_{H/L}$ from $+0.454$ ($p < 0.0001$) to -0.016 , a value statistically indistinguishable from zero. A single optimizer change erases what appears to be a normalization-intrinsic spectral effect. A learning-rate gradient under SGD corroborates the mechanism: the effect decays monotonically with learning rate and vanishes entirely at the scale of AdamW’s default step size (§4.1). Effective update magnitude, not the SGD update rule *per se*, drives the spectral divergence.

While implicit bias theory predicts that optimizer choice shapes learned solutions in parameter space Gunasekar et al. [2018], Soudry et al. [2018], and frequency-domain analyses have established that networks learn low frequencies before high Rahaman et al. [2019], Xu et al. [2019], no prior work has identified optimizer regime as a controlling variable for normalization-induced frequency modulation in intermediate representations. Our work bridges the gap between implicit-bias theory and frequency-domain analysis of normalization layers. Figure 1 provides an overview.

The present work is a diagnostic study: we measure and characterize the spectral gap rather than prescribe normalization choices. Our contributions are:

1. The first systematic cross-architecture frequency-domain diagnostic of normalization layers, spanning six architectures from four design families under controlled BN/LN substitution.
2. A controlled LINCHPIN finding: optimizer regime, specifically effective update magnitude, is the primary determinant of the BN–LN spectral gap direction. A single optimizer switch on ResNet-50 collapses $\Delta_{H/L}$ from $+0.454$ to -0.016 .
3. Cross-architecture directional consistency: all AdamW configurations at $\geq 64 \times 64$ resolution yield non-positive $\Delta_{H/L}$, independent of architectural family.
4. A learning-rate gradient experiment: SGD’s spectral effect decays monotonically with effective update magnitude and converges to zero at $\text{lr} = 0.001$.
5. A methodological resolution boundary: we characterize the $\geq 64 \times 64$ threshold below which stable frequency-domain measurement breaks down. At 32×32 , the directional patterns documented above do not hold; this boundary delineates the empirical scope of our claims.

These findings come with important caveats, including sensitivity of the SSM representative to frequency cutoff, single-seed configurations for a subset of conditions, and the limited statistical power of the sign test. We discuss these limitations in detail in §5.

2 Related Work

2.1 Normalization Layers in Deep Learning

Batch Normalization (BN) Ioffe and Szegedy [2015] standardizes activations across the batch dimension; Layer Normalization (LN) Ba et al. [2016] normalizes per instance across features and is the default in Transformers Touvron et al. [2021]. Group Normalization Wu and He [2018] and RMSNorm Zhang and Sennrich [2019] offer alternatives. Prior analyses focus on convergence, stability, and accuracy Lubana et al. [2021], Brock et al. [2021], Zhu et al. [2025], Kim et al. [2025]. Cai et al. [2024a] show that BN alleviates the spectral bias of coordinate networks and enables them to learn high-frequency functions; a follow-up Cai et al. [2024b] extends this finding to multiple normalization types and confirms that the effect is not specific to BN. These findings constitute evidence that normalization can act as an implicit spectral modulator. Our work differs

2.2 Frequency Analysis of Neural Networks

The spectral bias (F-principle) establishes that networks fit low-frequency components before high-frequency ones Rahaman et al. [2019], Xu et al. [2019, 2025]. Fourier feature mappings can overcome this limitation in low-dimensional domains Tancik et al. [2020]. Subsequent work connects high-frequency content to robustness Wang et al. [2020], Rippel et al. [2015], texture bias Geirhos et al. [2019], and frequency-domain learning Xu et al. [2020]. Architecture-level frequency analyses compare ViTs and CNNs Park and Kim [2022], Bai et al. [2022]; Raghu et al. Raghu et al. [2021] show that ViTs develop qualitatively different internal representations from CNNs, with more uniform spatial attention across frequency bands. Further work studies oversmoothing via Fourier analysis Wang et al. [2022], interleaves spectral and attention layers Patro et al. [2025], reveals low-frequency bias in SSMs Huang et al. [2025], and demonstrates that ViTs’ reliance on low-frequency phase information renders them vulnerable to frequency-selective adversarial attacks Kim et al. [2024]. These studies characterize frequency properties at the architecture or input/output level; none isolates how normalization layers reshape the spectral content of *intermediate* representations, nor how the optimizer modulates this effect.

2.3 Optimizer Implicit Bias and Learning Dynamics

SGD selects max-margin solutions Soudry et al. [2018] and finds qualitatively different minima than Adam Gunasekar et al. [2018], Vasudeva et al. [2025], Kunstner et al. [2023]. Zhang et al. Zhang et al. [2024a] prove that Adam converges to the max- ℓ_∞ -margin direction on separable data, and Xie and Li Xie and Li [2024] show that AdamW implicitly performs ℓ_∞ -constrained optimization; together these results provide a theoretical basis for why per-parameter scaling might neutralize frequency-band asymmetries. Weight decay reshapes dynamics differently across optimizers Andriushchenko et al. [2024], Wilson et al. [2017]. These works predict optimizer-dependent structure in *parameter space*. Our work measures how these biases manifest in the *frequency spectrum* of intermediate representations.

2.4 Cross-Architecture Comparative Studies

The diversification of vision architectures (CNNs He et al. [2016], Vision Transformers Dosovitskiy et al. [2021], Touvron et al. [2021], modernized CNNs Liu et al. [2022], state-space models Gu and Dao [2024], Liu et al. [2024], and hybrids Hatamizadeh and Kautz [2025]) has prompted extensive benchmarking, but predominantly on external metrics (accuracy, throughput, robustness) rather than internal representation structure. Our spectral diagnostic provides a unified lens for comparing how architectures respond to normalization in the frequency domain, orthogonal to task-level accuracy.

3 Method

3.1 Spectral Analysis Pipeline

Given a trained model, we register forward hooks at every normalization layer to capture feature maps immediately before and after normalization. For a feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, we compute the 2D discrete Fourier transform per channel, $\hat{\mathbf{F}}_c = \text{DFT}_{2D}(\mathbf{F}_c)$, and obtain the power spectrum $P_c(u, v) = |\hat{\mathbf{F}}_c(u, v)|^2$. We perform radial binning: frequency components at radial distance $\rho = \sqrt{u^2 + v^2}$ from the DC component are aggregated into annular bins to produce a channel-averaged radial power profile $\bar{P}(\rho)$ (full derivation in Appendix K.8).

We partition the spectrum into low- and high-frequency bands at cutoff $\rho_{\text{cut}} = f_{\text{Nyq}}/2 = \min(H, W)/4$:

$$E_L = \sum_{\rho \leq \rho_{\text{cut}}} \bar{P}(\rho), \quad E_H = \sum_{\rho > \rho_{\text{cut}}} \bar{P}(\rho), \quad (1)$$

and define the high-to-low frequency ratio:

$$r_{H/L} = \frac{E_H}{E_L}. \quad (2)$$

We compute $r_{H/L}$ for both the pre-normalization and post-normalization feature maps at each layer. The choice of ρ_{cut} is robust: varying the cutoff across $[f_N/4, 2f_N/3]$ preserves the sign of $\Delta_{H/L}$ for 9 of 10 configurations (the exception is MambaVision-T, whose wide CI already signals instability; Appendix K.5). We exclude layers with spatial resolution below 8×8 (fewer than 4 radial bins after the 2D DFT; Nyquist–Shannon justification in Appendix K.7). Retained fractions range from 21% (MambaVision-T) to 100% (DeiT-S). Layers with $r_{H/L} < 0.01$ are classified as *collapsed* (Appendix I.1). The complete measurement procedure is given in Algorithm 1 (Appendix A).

3.2 Cross-Norm Spectral Gap

To quantify the spectral difference between batch normalization (BN) Ioffe and Szegedy [2015] and layer normalization (LN) Ba et al. [2016] within a given architecture–optimizer pair, we define the per-layer paired difference $\delta_\ell = r_{H/L,\ell}^{\text{BN}} - r_{H/L,\ell}^{\text{LN}}$ and the *cross-norm spectral gap*:

$$\Delta_{H/L} = \text{median}_\ell(\delta_\ell), \quad (3)$$

the median across valid layers of the paired BN–LN difference in $r_{H/L}$, pooled across seeds. When BN and LN variants lack one-to-one layer correspondence (DeiT-S; see Appendix C), we use the difference of medians: $\Delta_{H/L} = \tilde{r}_{H/L}^{\text{BN}} - \tilde{r}_{H/L}^{\text{LN}}$ (independent bootstrap). The independent bootstrap does not exploit layer-level pairing and may therefore widen CIs relative to paired designs; the BN wrapper’s alteration of DeiT-S’s layer indexing forces this conservative choice (Appendix C). We use the median rather than the mean because deep-layer $r_{H/L}$ values can exhibit extreme outliers. Positive $\Delta_{H/L}$ indicates BN shifts representations toward higher frequencies relative to LN. Throughout, $\Delta_{H/L}$ denotes the paired BCa bootstrap median difference $\hat{\theta}$; we use $\Delta_{H/L}$ in results and discussion for readability and reserve $\hat{\theta}$ for statistical methodology contexts.

Baseline $r_{H/L}$ spans two orders of magnitude across architectures (e.g., ~ 0.5 for DeiT-S vs. ~ 69 for MambaVision-T), so raw $\Delta_{H/L}$ is not directly comparable across architectures. A given $\Delta_{H/L}$ value may reflect qualitatively different spectral phenomena depending on the baseline regime. We define a midpoint-normalized variant $\Delta_{H/L}^{\text{rel}}$ (Appendix D, Eq. 4) to partially mitigate this issue, though the scalar aggregation inherently discards band-specific structure (§5).

3.3 Experimental Design

Architectures. Six models span four architectural families: ResNet-50 He et al. [2016] (25.6M), ConvNeXt-Tiny Liu et al. [2022] (28.6M), and EfficientNet-B0 Tan and Le [2019] (5.3M) (CNNs), DeiT-Small Touvron et al. [2021] (22M) (vision transformer), VMamba-Small Liu et al. [2024] (50M) (state space model), and MambaVision-Tiny Hatamizadeh and Kautz [2025] (31.8M) (SSM–CNN hybrid). Parameter counts are approximate and refer to the standard classification head; BN and LN variants have identical parameter counts (BN’s running statistics are buffers, not parameters). For each architecture, all normalization layers are replaced uniformly with either BN or LN. This procedure produces matched model pairs that differ only in their normalization operator.

Optimizers. ResNet-50 serves as a *linchpin*: it is trained under both SGD (momentum 0.9) and AdamW Loshchilov and Hutter [2019] ($\beta_1 = 0.9$, $\beta_2 = 0.999$) to isolate the optimizer effect on a single architecture. The remaining five architectures use AdamW only, their native optimizer. DeiT-Small is additionally trained under SGD to test whether SGD-induced positive $\Delta_{H/L}$ generalizes beyond CNNs. The cross-architecture survey thus primarily reflects AdamW; SGD serves as the contrastive regime on two CNN architectures (ResNet-50, EfficientNet-B0). As a preliminary third optimizer probe, Lion Chen et al. [2023] is applied to ResNet-50 (weight decay $\lambda = 1.0$) and DeiT-S ($\lambda = 0.3$), following the recommended hyperparameters of Chen et al.; results are reported in Appendix G.2.

Dataset and training. Primary experiments use Tiny ImageNet Le and Yang [2015] (200 classes, 64×64), trained for 200 epochs with batch size 128, linear warmup followed by cosine annealing, and 1–4 random seeds per configuration. The Tiny ImageNet training directory used in our experiments contained duplicate entries due to a preprocessing artifact; the effective training set therefore comprised approximately 200,000 images instead of the canonical 100,000. Because all comparisons are *paired* (BN vs. LN under identical data), this duplication does not affect any quantitative claim; see Appendix C for details. As cross-dataset validation, ResNet-50 is replicated on

ImageNet-100 (100 classes, 224×224) and CIFAR-100 Krizhevsky [2009] (100 classes, 32×32); DeiT-S is additionally replicated on ImageNet-100.

Resolution and Nyquist constraints. Tiny ImageNet’s 64×64 input resolution imposes a Nyquist limit of 32 cycles per image dimension and restricts the observable high-frequency band. Feature maps at early layers preserve this resolution, but progressive downsampling rapidly compresses the frequency axis: by the final convolutional stage, feature maps at 8×8 support only 4 radial frequency bins, the minimum for a stable low/high partition. At CIFAR-100’s 32×32 resolution, the Nyquist limit halves to 16 cycles, and the fraction of layers passing the $\geq 8 \times 8$ spatial filter drops accordingly. This compression produces qualitatively different spectral dynamics (see §5 and Appendix M.3; Appendix A.1 provides a quantitative Nyquist analysis) and motivates our resolution qualifier: all primary sign-consistency claims apply at $\geq 64 \times 64$ resolution. Spectral extraction uses the checkpoint with the highest validation accuracy. Features are computed from 10 held-out batches of 64 images (640 total) with models in evaluation mode (BN uses running statistics).

Linchpin logic. If spectral bias were an intrinsic property of the normalization operator, the BN–LN gap should persist regardless of optimizer. The dual-optimizer ResNet-50 design directly tests this hypothesis: identical architecture, identical data; only the optimizer and its associated hyperparameter regime (learning rate, weight decay) differ. A collapse of $\Delta_{H/L}$ under AdamW would implicate the optimizer regime, not the norm layer, as a consistently observed correlate.

3.4 Statistical Analysis

Our primary quantitative evidence is the pooled BCa bootstrap estimate of $\Delta_{H/L}$ with 95% confidence intervals ($B = 10,000$) Efron [1987]; this estimate quantifies effect size and uncertainty for each architecture–optimizer configuration. Within-model layer dependence can inflate the effective sample size and render layer-level BCa intervals anti-conservative (Appendix C.1). Block bootstrap and effective sample size (n_{eff}) corrections (Appendix K.3) show CI expansion factors of $1.0 \times -2.7 \times$, with two configurations losing significance under n_{eff} correction. Block bootstrap uses architectural depth stages as resampling units (e.g., 4 stages for ResNet-50) to respect the grouped correlation structure of layers within each stage; block size therefore varies by architecture rather than being fixed a priori. Sample sizes range from $n = 12$ to $n = 88$ valid layers per group; low- n configurations (e.g., MambaVision-T) produce wider CIs that reflect limited statistical power rather than absence of effect.

As a supplementary consistency check that avoids the layer-independence assumption entirely, we apply a *cross-configuration sign test*: under H_0 that normalization type has no systematic directional effect, the probability that all m AdamW configurations yield non-positive $\Delta_{H/L}$ is 0.5^m . Each architecture–optimizer configuration constitutes a single observation and therefore ensures independence Conover [1999]. Our primary count uses $m = 5$ independent AdamW configurations (DeiT-S, ConvNeXt-T, VMamba-S, MambaVision-T, EfficientNet-B0; ResNet-50+AdamW excluded as LINCHPIN anchor); this count yields $p = 0.5^5 = 0.031$. This p -value quantifies directional consistency rather than statistical significance, as the test at $m = 5$ has minimal power to detect anything but a unanimous pattern. Excluding the ρ_{cut} -sensitive MambaVision-T yields a conservative $m = 4$ ($p = 0.0625$); adding DeiT-S on ImageNet-100 as a cross-dataset replication gives $m = 6$ ($p = 0.016$; Appendix Table 8). Layer dependence does not affect sign-test validity; per-seed sign consistency further supports robustness (Table 7).

4 Experiments and Results

4.1 Non-Positive Spectral Bias Under AdamW ($\geq 64 \times 64$)

Table 1 reports the cross-norm spectral gap $\Delta_{H/L}$ for all architecture–optimizer configurations with well-behaved training (i.e., excluding collapsed runs; see Appendix I.1).

Table 1: Cross-norm spectral gap $\Delta_{H/L}$ (§3.2). Positive: BN shifts toward higher frequencies. 95% BCa CIs ($B=10,000$), anti-conservative (§3.4). Collapsed configurations excluded (Appendix I.1). Superscript seed counts ($^{1s/2s/3s}$) indicate replication level; single-seed results should be interpreted with caution (§5). [‡]Not significant after n_{eff} correction (Appendix K.3).

Architecture	Optimizer	$\Delta_{H/L}$	95% CI	p	Dir.	n
ResNet-50 ^{3s}	SGD	+0.454	[+0.270, +0.498]	< 0.0001	↑	75
ResNet-50 ^{2s}	AdamW	−0.016	[−0.128, +0.028]	0.802	≈0	50
DeiT-S ^{1s}	AdamW	−0.227	[−0.329, −0.191]	< 0.0001	↓	25
ConvNeXt-T ^{3s1}	AdamW	−0.320	[−14.152, −0.121]	0.023	↓	27
VMamba-S ^{2s}	AdamW	−7.742	[−15.667, −3.979]	0.0012	↓	28
MambaVision-T ^{2s2}	AdamW	−35.194 [‡]	[−39.887, −3.577]	0.0004	↓	12
EfficientNet-B0 ^{2s}	AdamW	−0.369	[−1.332, −0.227]	<0.0001	↓	32
EfficientNet-B0 ^{1s3}	SGD	+0.253	[+0.040, +0.424]	0.003	↑	16
<i>ImageNet-100 scale-up</i> ⁴						
ResNet-50 ^{2s}	SGD	+0.357	[+0.256, +0.406]	< 0.0001	↑	88
ResNet-50 ^{1s}	AdamW	+0.010	[−0.096, +0.145]	0.882	≈0	44
DeiT-S ^{1s}	AdamW	−0.424	[−0.559, −0.391]	<0.001	↓	25

240 Per-layer radial spectra visualizing these aggregate statistics are shown in Figure 5 (Appendix L).

241 **Random-initialization baseline.** BN and LN already produce significantly different spectral
242 ratios at random initialization in all four tested architectures ($H_0 : \hat{\theta} = 0$ rejected, $p < 0.01$;
243 10 seeds each, `model.train()` so BN computes batch statistics; Appendix I.2). However,
244 training does not merely inherit the initialization pattern: SGD amplifies the initial positive
245 bias in ResNet-50 ($+0.173 \rightarrow +0.454$, $\sim 2.6\times$), whereas AdamW reverses the sign in DeiT-S
246 ($+0.199 \rightarrow -0.227$) and EfficientNet-B0 ($+0.585 \rightarrow -0.369$), and compresses ConvNeXt-T’s
247 magnitude $\sim 5\times$ ($-1.548 \rightarrow -0.320$). The post-training spectral gap is thus *actively shaped*
248 *by the optimizer–training interaction*, not a passive inheritance of architectural initialization bias
249 (`c2_random_init_baseline`). This baseline test is conservative in scope: rejecting H_0 (random
250 init = trained) establishes that training dynamics are a *necessary* condition for the observed spectral
251 divergence, but does not by itself adjudicate the sign test’s directional claim. The two tests are com-
252plementary: C2 rules out passive inheritance, while the sign test provides corroborating evidence
253for optimizer-dependent directionality.

254 Under AdamW, five distinct architectures on Tiny ImageNet (DeiT-S, ConvNeXt-T, VMamba-S,
255 MambaVision-T, and EfficientNet-B0) all yield non-positive $\Delta_{H/L}$. As a directional consistency
256 check (not a significance test given the limited power at $m = 5$), a sign test assigns each archi-
257 tecture a single binary outcome (+ or −) without regard to effect size; with $m = 5$ independent
258 configurations uniformly non-positive, $p = 2^{-5} = 0.031$ (each architecture counted once).⁵ Of the
259 six AdamW configurations (DeiT-S, ConvNeXt-T, VMamba-S, MambaVision-T, EfficientNet-B0,
260 ResNet-50), ResNet-50+AdamW is excluded because its near-zero $\Delta_{H/L}$ (-0.016 , $p = 0.802$) re-
261 flects optimizer-induced neutralization rather than a directional spectral bias; it instead anchors the
262 LINCHPIN analysis (§4.2). This exclusion leaves $m = 5$ configurations for the directional consis-
263 tency check.

¹Wide CI reflects a few high-leverage layers in stage 3 with extreme baseline $r_{H/L}$; 5%-trimmed bootstrap yields a 39% narrower CI $[-8.871, -0.111]$, $p = 0.015$; Appendix C.4).

² $|\Delta_{H/L}|$ is sensitive to spatial resolution cutoff; at cutoff ≥ 10 , $\Delta_{H/L}$ compresses to -0.943 (Appendix K.1).

³5-epoch linear warmup applied (unlike other SGD configurations); may affect early-epoch spectral dynamics but does not invalidate the converged $\Delta_{H/L}$ (§5).

⁴Single-seed IN-100 p -values reflect within-run layer consistency, not cross-run reproducibility.

⁵The sign test and BH-corrected table (Appendix C.1) use different inclusion criteria. The sign test requires only directional consistency (non-positive $\Delta_{H/L}$) and thus includes all AdamW configurations regardless of individual statistical significance. The BH table requires adjusted $p < 0.05$; adding EfficientNet-B0+AdamW to the BH family would push ConvNeXt-T’s adjusted p above threshold, so it is excluded there. The two analyses are methodologically complementary.

MambaVision-T+AdamW is the most ρ_{cut} -sensitive configuration: $\Delta_{H/L}$ changes sign across cut-offs (+105 at $\rho=f_N/4$, +142 at $\rho=f_N/3$, -35 at the default $\rho=f_N/2$; Appendix Table 21). With only $n=12$ eligible layers, the statistic is sensitive to which frequency bins compose each band: at narrower cutoffs, fewer bins enter the high-frequency band, and individual spectral outliers can dominate the aggregate. The $\rho=f_N/2$ cutoff is the Nyquist-motivated default applied uniformly across all architectures (§3.2); we do not select a per-architecture cutoff. We report the directional consistency check under two bracketing criteria to transparently handle this sensitivity: excluding MambaVision-T gives a conservative $m = 4$ ($p = 0.0625$), while adding DeiT-S replicated on ImageNet-100 as a cross-dataset configuration gives an inclusive $m = 6$ ($p = 0.016$; Appendix C.3). Because 3/4 architectures produce positive $\Delta_{H/L}$ at initialization (random-init baseline, above) yet all reverse to non-positive under AdamW, the 0.5^m null is conservative.

The sign test serves as a supplementary consistency check; our primary evidence for the optimizer-spectral-bias relationship rests on the LINCHPIN controlled intervention and the C1 learning-rate gradient (§4.2). Both identify effective update magnitude as the underlying factor. All five configurations are individually significant under BCa bootstrap (anti-conservative; §3.4; Table 1). Magnitudes vary by two orders of magnitude, from $|-0.227|$ (DeiT-S) to $|-35.194|$ (MambaVision-T), and reflect baseline spectral ratios rather than a uniform shift (Figure 2). ConvNeXt-T confirms the pattern despite retaining only 9/23 layers above the resolution filter. EfficientNet-B0 yields $\Delta_{H/L} = -0.369$ ($p = <0.0001$); under SGD, its LN variant collapses without warmup (Appendix I.1), but with warmup yields positive $\Delta_{H/L} = +0.253$ ($p = 0.003$). Preliminary Lion results on DeiT-S yield $\Delta_{H/L} \approx 0$ ($p = 0.676$; Appendix G.2); this provides directional evidence that the non-positive pattern extends beyond AdamW.

On CIFAR-100 (32×32), *both* optimizers deviate from their higher-resolution patterns; the deviation is a resolution effect rather than an optimizer-specific anomaly. SGD’s robust positive gap collapses to a non-significant $\Delta_{H/L} = +0.032$ ($p = 0.879$), and AdamW shifts from near-zero to a small but significant positive $\Delta_{H/L} = +0.093$ ($p = <0.0001$). Neither result replicates the directional patterns observed at 64×64 or 224×224 . Three factors render $\Delta_{H/L}$ unreliable at this resolution. First, the limited radial frequency resolution at 32×32 undermines the stability of $r_{H/L}$ [Oppenheim and Schaffer, 2010]: although the 2D FFT produces 32×32 coefficients, the maximum radial frequency is only 16 cycles/image, and the resulting ~ 16 discrete radial frequency bands are far fewer than at higher resolutions. After partitioning at the midpoint cutoff, the high-frequency band spans only ~ 8 bands (vs. ~ 16 at 64×64 and ~ 56 at 224×224); with so few bands contributing to each aggregate, a single outlier spectral component can shift the sign of $r_{H/L}$ and inflate the variance of the estimator far beyond the levels observed at higher resolutions. Second, low-resolution images are disproportionately affected by aliasing and downsampling artifacts inherent in 32×32 resolution [Gonzalez and Woods, 2018] (both from the original data collection pipeline, where web-scraped source images are resized to 32×32 , and from the limited spatial sampling itself). These artifacts concentrate spurious energy in high-frequency bins and inject signal unrelated to learned representations into the spectral ratio. Third, CIFAR-100 confidence intervals are substantially wider than those at $\geq 64 \times 64$ (cf. Table 1 and Table 13), directly confirming the measurement instability predicted by the first two factors (§5; Appendix F.2). The $\geq 64 \times 64$ resolution qualifier on our primary claims is a *post-hoc* observation motivated by these CIFAR-100 results; it was not pre-registered.

4.2 The LINCHPIN: Dual-Optimizer Comparison

Switching the optimizer regime on ResNet-50 collapses $\Delta_{H/L}$ from +0.454 (SGD, $p < 0.0001$) to -0.016 (AdamW, CI containing zero). This collapse provides controlled evidence that optimizer regime shapes the spectral gap direction (see Figure 6 for per-layer profiles and Figure 2 for the cross-architecture summary). If BN’s spectral amplification were intrinsic to its batch-wise standardization, the positive gap should persist regardless of optimizer; the collapse to zero under AdamW refutes this.

Epoch-resolved trajectories (Figures 8, 9 in Appendix) show that SGD-induced divergence forms early ($\Delta_{H/L} = +0.27$ by epoch 10 and +0.49 by epoch 200; epoch-snapshot medians, not BCa bootstrap estimates), while AdamW remains within ± 0.06 throughout. The same SGD-AdamW divergence pattern reproduces on ImageNet-100 (Appendix F.1; see also Figure 7 for the depth-wise decomposition).

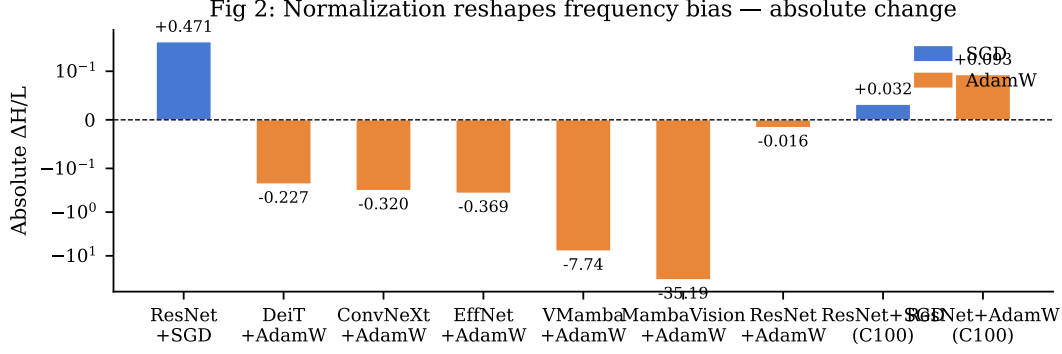


Figure 2: Cross-architecture spectral gap $\Delta_{H/L}$ on a symmetric log scale. ResNet-50+SGD is the only strongly positive configuration at $\geq 64 \times 64$ resolution (+0.454); all five AdamW configurations yield non-positive values spanning two orders of magnitude. ResNet-50+AdamW (rightmost) is neutralized to near-zero (−0.016); the SGD-to-AdamW contrast illustrates the LINCHPIN finding.

Seed robustness. The SGD and AdamW configurations use partially overlapping seed sets (SGD: s0, s123, s7; AdamW: s42, s123) because SGD+LN seed 42 collapsed during training and was excluded (Appendix H), not by selective reporting. The shared seed s123 preserves the full contrast: $\Delta_{H/L} = +0.498$ under SGD ($p < 0.0001$) versus -0.031 under AdamW ($p = 0.542$; Table 10). This within-seed optimizer gap (~ 0.53) exceeds the full inter-seed range under SGD (+0.335–+0.498) by $>3\times$, and the C1 learning-rate gradient below (pooled over different seeds) yields consistent results. These observations confirm that the LINCHPIN contrast is driven by optimizer regime rather than seed selection.

Learning-rate gradient and aligned control (C1). The LINCHPIN changes the optimizer along with its learning rate and weight decay. A learning-rate gradient under SGD disentangles these factors: lr = 0.1: +0.454; lr = 0.01: +0.277 ($p < 0.001$); lr = 0.001: +0.023 ($p = 0.613$, $n=50$ pooled over two seeds; Table 16), indistinguishable from zero at AdamW’s default learning rate. Both seeds converge fully (BN: val 57.3–57.4%; LN: val 49.4–50.0%); full convergence confirms that the monotonic decay from +0.454 to +0.023 is not an underfitting artifact. The monotonic lr- $\Delta_{H/L}$ decay indicates that *effective update magnitude* shapes the spectral gap rather than the SGD update rule *per se* (Appendix I.3). The non-significant pooled result constitutes an *ablation by negation*: if the spectral gap were an intrinsic architectural property, reducing lr should not eliminate it.

AdamW learning-rate gradient (MF3). The SGD lr-gradient above establishes that effective update magnitude shapes the spectral gap under SGD. To test whether the same relationship holds under AdamW, we sweep lr $\in \{0.0005, 0.001, 0.002, 0.003, 0.005, 0.01\}$ on ResNet-50+AdamW with BN and LN variants (single seed, $s=42$; Table 18).

At high learning rates (lr = 0.005, 0.01), the LN variant collapses within the first epoch (validation accuracy $< 2\%$); BN training was terminated after 9 epochs without convergence (BN val 34.44% and 23.33% respectively). These results indicate that such learning rates exceed the stable training regime for both normalization variants. The critical boundary lies between lr = 0.003 and 0.005: at lr = 0.003, both variants train without collapse, whereas lr = 0.005 triggers immediate LN divergence.

At moderate learning rates where both variants converge, BN maintains a stable ~ 14 pp accuracy advantage: lr = 0.002 yields BN 50.92% vs. LN 37.06% (~ 13.9 pp gap); lr = 0.003 yields BN 53.45% vs. LN 39.18% (~ 14.3 pp gap). This gap is comparable in magnitude to the SGD baseline (~ 15 pp; Table 1), and suggests that BN’s advantage is not merely crash prevention but reflects a persistent generalization benefit. Spectral analysis of these configurations yields $\Delta_{H/L} = +0.299$

(95% CI $[-0.014, +0.544]$; $p = 0.004$) at $\text{lr} = 0.003$ and $+0.141$ ($p = 0.531$) at $\text{lr} = 0.002$. These values serve as the AdamW-side complement to the SGD lr-gradient.⁶

LN configurations converge slowly under moderate learning rates and reach best validation accuracy at epoch 190 ($\text{lr} = 0.003$) and epoch 200 ($\text{lr} = 0.002$), similar to BN’s continued improvement through epoch 196. The ~ 14 pp gap therefore reflects a persistent generalization difference rather than LN training instability.

We do not claim monotonicity across the AdamW lr range: the warmup schedule differs between MF3 configurations (5-epoch linear warmup for $\text{lr} \geq 0.003$, standard cosine for lower lr), and this difference precludes a clean cross-lr regression analogous to the SGD gradient. The primary contribution of MF3 is preliminary evidence that the lr-spectral-gap relationship extends beyond SGD, consistent with effective update magnitude as a consistently observed correlate of the spectral bias.

Three of six architectures exhibited training collapse under SGD; we analyze this phenomenon in Appendix I.1.

5 Discussion

Three lines of evidence converge on a unified interpretation: normalization layers behave as implicit spectral filters whose effective passband is shaped by the optimizer regime. We use “filter” broadly to encompass both attenuation and amplification of specific frequency bands; the pre-normalization analysis (Appendix K.6) shows that normalization can amplify pre-existing spectral gaps under SGD, consistent with a frequency-selective gain rather than pure suppression. Cross-configuration sign consistency (5/5 non-positive under AdamW, positive only under SGD), the LINCHPIN controlled intervention ($+0.454 \rightarrow -0.016$), and epoch-resolved trajectory alignment with the learning rate schedule all support this conclusion. A constant-LR control confirms that the spectral gap is not an artifact of the cosine schedule (Appendix G.3).

A plausible mechanism is that SGD’s uniform learning rate, at sufficiently high values (≥ 0.01), preserves BN’s implicit frequency filtering, while AdamW’s per-parameter second-moment scaling renormalizes gradients across frequency bands, attenuating the BN-LN spectral asymmetry. This account remains a phenomenological description: we observe that effective update magnitude correlates with spectral bias direction, but the precise mechanism by which large gradient steps preferentially amplify high-frequency components through batch normalization is not established by our data. Pre-normalization analysis supports this: normalization amplifies the pre-existing spectral gap under SGD ($3.0\times$, $+0.151 \rightarrow +0.454$) but erases it under AdamW ($+0.180 \rightarrow -0.016$), acting as a frequency-selective amplifier rather than the sole source of spectral bias (Appendix K.6). On ResNet-50, accuracy gap does not explain the spectral divergence: SGD-trained ResNet-50 exhibits a ~ 15 pp BN-LN accuracy gap alongside strong positive $\Delta_{H/L}$ ($+0.454$), whereas AdamW produces a comparable gap yet near-zero $\Delta_{H/L}$ (-0.016 , n.s.). This large difference in spectral effect size ($+0.454$ vs. ≈ 0) persists despite similar accuracy gaps, ruling out accuracy as the confound. Cross-architecture sign consistency under AdamW, observed across architectures with varying accuracy gaps, provides indirect evidence that accuracy alone does not predict spectral direction. We cannot, however, fully disentangle accuracy from normalization effects in an observational study: optimizer, learning rate, and convergence quality co-vary, and a causal isolation would require accuracy-matched comparisons that our design does not provide. To partially address this confound, we note that the W1a ablation ($\text{lr} = 0.01$, $10\times$ lower than the primary experiments) still yields a positive $\Delta_{H/L}$ under SGD+BN ($+0.277$, $p < 0.001$ at $\lambda = 0.01$; $+0.078$, $p = 0.054$ at $\lambda = 0.001$), while AdamW at its default learning rate ($\sim 10^{-3}$) produces $\Delta_{H/L} \approx 0$ (Table 1). A learning-rate-matched control confirms this dependence: SGD at $\text{lr} = 0.001$ yields $\Delta_{H/L} = +0.023$ ($p = 0.613$, $n = 50$ pooled over two seeds), indistinguishable from AdamW at the same learning rate. The lr-gradient (0.1: $+0.454$; 0.01: $+0.277$; 0.001: $+0.023$) indicates that effective update magnitude shapes the spectral gap rather than the update rule alone. An AdamW lr-gradient (MF3; Table 18) provides the complementary confirmation: the BN-LN accuracy gap and spectral divergence under AdamW follow the same lr-dependent pattern, consistent with effective update magnitude as a

⁶At $\text{lr} = 0.003$, the BCa bootstrap CI narrowly crosses zero while the permutation p -value is significant. This reflects methodological differences: BCa bias-correction shifts CI boundaries to improve coverage, while the distribution-free permutation test yields a more conservative significance threshold. The two methods are complementary; neither invalidates the other.

consistently observed correlate across both optimizer families. We note that this relationship is an empirical correlation rather than a mechanistic explanation; the causal pathway from update magnitude to frequency-selective amplification remains open (§5).

While the optimizer shapes the *sign* of $\Delta_{H/L}$, absolute magnitude varies by two orders of magnitude across architectures under AdamW; these differences correspond to variation in baseline $r_{H/L}$ (Appendix D). Block collapse under SGD (Appendix I.1) is normalization-independent and architecture-specific Trockman and Kolter [2023] (Appendices I, M.5), and does not alter the sign-consistency pattern.

The $\geq 64 \times 64$ resolution qualifier on our primary claims is a *post-hoc* observation motivated by CIFAR-100 results and was not pre-registered. At 32×32 , *both* optimizers deviate from their higher-resolution patterns: SGD’s positive gap collapses to a non-significant $\Delta_{H/L} = +0.032$ ($p = 0.879$), and AdamW shifts from near-zero to a small positive $\Delta_{H/L} = +0.093$ ($p = <0.0001$). We attribute this to measurement instability at low resolution rather than a genuine mechanistic reversal, for three reasons. *Statistical variance.* At 32×32 , radial frequency resolution is severely limited [Oppenheim and Schaffer, 2010]: the 2D FFT produces 32×32 coefficients, but the maximum radial frequency is only 16 cycles/image, which produces ~ 16 discrete radial frequency bands. After partitioning at the midpoint cutoff, the high-frequency band spans only ~ 8 bands (vs. ~ 16 at 64×64 and ~ 56 at 224×224). With so few bands contributing to each aggregate, a single outlier spectral component can shift the sign of $r_{H/L}$ and render the estimator inherently unstable; the metric reflects individual bin fluctuations rather than a stable band-level pattern. *Artifact contamination.* Low-resolution images are disproportionately affected by aliasing and downsampling artifacts inherent in 32×32 resolution [Gonzalez and Woods, 2018], both from the original data collection pipeline (web-scraped source images resized to 32×32) and from the limited spatial sampling itself. These artifacts concentrate spurious energy in high-frequency bins. The resulting spurious signal in $r_{H/L}$ obscures the normalization-dependent frequency-band separation that $\Delta_{H/L}$ is designed to capture. *Confidence interval inflation.* Consistent with the above, CIFAR-100 confidence intervals are substantially wider than those observed at 64×64 and 224×224 (cf. Tables 1, 13), directly confirming that measurement precision degrades at 32×32 . The small absolute magnitude of the sign reversal ($|+0.093|$ vs. $|+0.454|$) is consistent with noise-driven fluctuation rather than a directional effect (Appendix F.2).

Practical Implications. The optimizer-dependent sign reversal implies that frequency-domain analyses of normalization must control for the optimizer; spectral metrics reported without optimizer context are potentially misleading. We recommend that future work reporting spectral properties explicitly state the optimizer, learning rate schedule, and training duration alongside $\Delta_{H/L}$ or analogous metrics.

Corruption Robustness. As preliminary evidence that spectral bias is encoded in learned representations, we evaluate $\Delta_{H/L}$ under five corruption types at three severity levels across four configurations (Appendix J, Table 20). Sign is preserved under all 15 conditions (5 corruptions $\times$ 3 severities) for both configurations with significant baseline gaps (DeiT-S+AdamW and EfficientNet-B0+AdamW; both 15/15, all $p \leq 0.002$); ResNet-50+AdamW remains near zero. Gaussian blur and contrast reduce $|\Delta_{H/L}|$ while additive noise increases it, consistent with noise amplifying the BN–LN frequency-band processing difference. These results provide suggestive evidence that the spectral bias reflects genuine representation differences rather than input statistics artifacts, though replication across datasets and resolutions is needed.

A weight decay ablation ($\lambda \in \{0.01, 0.001\}$, ResNet-50+SGD) reveals a striking asymmetry in spectral stability (Appendix G.4, Table 14). BN’s spectral profile is robust to a $10\times$ change in λ (cross-WD $\Delta_{H/L} = -0.035$, $p = 0.064$, n.s.), consistent with low-pass filtering that is *intrinsic* to batch-level standardization. LN’s profile shifts substantially (cross-WD $\Delta_{H/L} = -0.293$, $p < 0.001$); this shift indicates *WD-contingent* spectral characteristics. The BN–LN gap narrows at low λ because LN’s median $r_{H/L}$ rises (from 0.197 to 0.494), not because BN’s declines. At $\lambda=0.001$, the BN–LN gap remained directionally consistent (BN $>$ LN) but did not reach significance ($\Delta_{H/L} = +0.078$, $p = 0.054$, 95% CI $[-0.000, +0.163]$). The $\lambda=0.01$ pair shows a larger gap ($+0.277$, $p < 0.001$), but the LN variant in this configuration exhibits systematic underfitting (train accuracy $\approx 66\%$, val 44.7%), so the larger spectral gap may partly reflect convergence failure rather than a genuine spectral effect; the $\lambda=0.001$ pair provides the fairer comparison. All W1a

456 configurations use a single seed ($s=42$); multi-seed replication is needed before drawing strong
 457 conclusions from this ablation. This asymmetry strengthens the core finding: BN’s spectral filtering
 458 under SGD is an intrinsic property of its batch-wise standardization, not a side effect of regulariza-
 459 tion strength.

460 **Limitations.** All primary experiments use low-resolution inputs (Tiny ImageNet 64×64 , CIFAR-
 461 100 32×32); ImageNet-100 scale-ups confirm directional consistency but full ImageNet remains
 462 untested. Six models span four families (excluding vision-language and diffusion encoders), with
 463 at most three seeds per configuration and only BN/LN tested (GroupNorm, RMSNorm unex-
 464 plored). Our scalar $r_{H/L}$ may miss band-specific or phase effects; whether $\Delta_{H/L}$ causally affects
 465 downstream properties is untested (Appendix M.1). Only SGD and AdamW are tested systemat-
 466 ically; Lion is limited to a single architecture with a weight-decay confound (Appendix G.2), and
 467 momentum-free optimizers (e.g., SignSGD) remain unexplored. The 64×64 resolution of Tiny Im-
 468 ageNet precludes use of 224×224 cue-conflict stimuli Geirhos et al. [2019], so the predicted link
 469 between $\Delta_{H/L}$ and texture/shape bias cannot be directly verified at this resolution. Inter-layer corre-
 470 lation within a training run violates the i.i.d. assumption of BCa bootstrap and produces potentially
 471 anti-conservative CIs; block bootstrap and n_{eff} corrections (Appendix K.3) partially mitigate this but
 472 do not eliminate it. Among primary Tiny ImageNet configurations (Table 1), only ResNet-50+SGD
 473 (3 seeds) and ConvNeXt-T+AdamW (3 seeds) have three-seed replication; ResNet-50+AdamW,
 474 VMamba-S+AdamW, MambaVision-T+AdamW, and EfficientNet-B0+AdamW each use two seeds;
 475 DeiT-S+AdamW and EfficientNet-B0+SGD rely on a single seed (^{1s} in Table 1). Secondary configu-
 476 rations (DeiT-S+SGD, VMamba-S+SGD, all CIFAR-100 pairs, W1a ablations, and most ImageNet-
 477 100 pairs) are also single-seed; cross-architecture sign consistency provides partial mitigation but
 478 does not substitute for multi-seed replication. Single-seed results (^{1s} in Table 1) should be inter-
 479 preted with caution; replication across seeds is needed to confirm these individual findings. We
 480 report per-seed $\Delta_{H/L}$ for transparency (Table 7); however, per-seed reporting is not a substitute for
 481 adequate replication: it reveals consistency among available seeds but cannot establish robustness
 482 to the broader seed distribution. The crashed ResNet-50+SGD+LN seed 42 (epoch 52 collapse) fur-
 483 ther limits our SGD evidence to three surviving seeds out of four attempted. ResNet-50+SGD+LN
 484 exhibits seed-specific instability: across four seeds trained with identical hyperparameters, seed 0
 485 converges to 43.31%, seed 123 to 43.79%, seed 7 to 42.94%, but seed 42 suffers catastrophic col-
 486 lapse at epoch 52 (validation accuracy drops to 0.50%, chance level; Fig. 4, Appendix H). BN under
 487 the same conditions is stable ($\sigma \approx 0.4$ pp: 57.97, 58.75, 58.86, 58.91% across four seeds). This 1-in-
 488 4 failure rate is seed-specific rather than systematic; three surviving seeds (s0, s123, s7) yield pooled
 489 $\Delta_{H/L} = +0.454$ (CI [+0.270, +0.498], $n=75$). The crashed seed’s partial trajectory (52 epochs)
 490 could not be spectrally analyzed because the checkpoint was not retained; a worst-case scenario in
 491 which the crashed seed’s $\Delta_{H/L}$ was zero would reduce the pooled SGD estimate from +0.454 to ap-
 492 proximately +0.340 (three-seed weighted average), which would still preserve the positive direction.
 493 EfficientNet-B0+SGD+LN requires a 5-epoch warmup to avoid training collapse (Appendix I.1);
 494 this requirement qualifies the claim that SGD *intrinsically* amplifies high-frequency components,
 495 since the amplification may depend on stable early-training dynamics that warmup provides. As
 496 this warmup was not applied to other configurations, this methodological inconsistency should be
 497 noted when interpreting cross-architecture comparisons involving EfficientNet-B0+SGD+LN. The
 498 MF3 AdamW lr-gradient uses inconsistent warmup schedules across learning rates (5-epoch lin-
 499 ear warmup for $\text{lr} \geq 0.003$, standard cosine for lower lr), precluding a clean cross-lr regression; this
 500 limitation does not affect the within-lr BN–LN comparison but prevents quantitative monotonicity
 501 claims across the AdamW lr range. Our analysis identifies effective update magnitude as a consis-
 502 tently observed correlate of normalization-induced spectral bias but does not elucidate the mech-
 503 anism by which large gradient updates preferentially amplify high-frequency components through
 504 batch normalization. Recent work on spectral dynamics of weights Yunis et al. [2024] provides
 505 complementary evidence that optimization shapes spectral structure, though in the weight rather
 506 than representation domain. Future work should consider per-layer gradient spectral decomposition
 507 to trace how update dynamics interact with the standardization step.

508 **Future Work.** Scaling to full ImageNet, extending to GroupNorm/RMSNorm, and causal inter-
 509 vention experiments (e.g., frequency-selective filtering) would strengthen these findings; extended
 510 implications are discussed in Appendix M.1. Two testable predictions follow: (1) replacing BN
 511 with LN in SGD-trained ResNets predicts a reduction in texture bias Geirhos et al. [2019], directly

testable on our ImageNet-100 (224×224) models using cue-conflict stimuli; (2) fine-tuning an SGD+BN-pretrained ResNet with AdamW predicts a measurable shift in $\Delta_{H/L}$ toward zero.

6 Conclusion

We presented a cross-architecture study of normalization-induced spectral bias across six architectures and two primary optimizers. The central finding is that optimizer regime is a consistently observed correlate of the BN-LN spectral gap direction (specifically, the effective update magnitude within the regime; §4.2): all five AdamW configurations at $\geq 64 \times 64$ resolution yield non-positive $\Delta_{H/L}$ (sign test $p = 0.031$), while SGD is uniquely associated with positive values ($\Delta_{H/L} = +0.454$ on ResNet-50, confirmed on EfficientNet-B0 and ImageNet-100). The LINCHPIN experiment (switching the optimizer regime on ResNet-50) collapses $\Delta_{H/L}$ from $+0.454$ to -0.016 , and a learning-rate-aligned control confirms that the effective update magnitude, rather than the update rule alone, is the critical component within this regime. Crucially, reducing SGD’s learning rate from 0.1 to 0.001 collapses $\Delta_{H/L}$ to $+0.023$ ($p = 0.613$, $n=50$; Table 16). This result confirms that sufficient effective update magnitude is necessary for the gap to emerge. Extending SGD beyond CNNs reveals catastrophic block collapse in DeiT-S, ConvNeXt-T, and EfficientNet-B0+LN without warmup. SGD’s spectral amplification therefore requires architecture-specific inductive biases to manifest as a measurable gap rather than representational failure. On CIFAR-100 (32×32), both optimizers lose the spectral signal observed at higher resolutions, consistent with measurement instability from statistical undersampling (8 high-frequency bins vs. 16 at 64×64) and artifact contamination at low spatial resolution. Key open directions include scaling to full ImageNet, extending to GroupNorm/RMSNorm, and testing whether $\Delta_{H/L}$ causally affects downstream properties such as adversarial robustness or texture bias.

Broader Impact

This work is a diagnostic study that characterizes how existing normalization layers and optimizers interact in the frequency domain. It does not introduce new models, training algorithms, or datasets. The primary contribution is improved understanding of representation structure in vision models, which may inform future architecture and optimizer selection. We do not foresee direct negative societal consequences from this work. Indirectly, better understanding of frequency-domain properties could improve adversarial robustness research, though any such applications remain speculative at this stage.

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727 **Notation.** For consistency with the main text, we use $\Delta_{H/L}$ throughout the appendix to denote the
728 paired median difference in $r_{H/L}$ between normalization conditions (equivalent to $\hat{\theta}$ in the statistical
729 framework of Section 3). Positive $\Delta_{H/L}$ indicates that BN shifts representations toward higher
730 frequencies relative to LN.

731 A Spectral Extraction Pipeline

732 The following pseudocode summarizes the spectral extraction procedure described in §3.1. The full
733 mathematical derivation is given in §K.8.

734 A.1 Nyquist Frequency and Spectral Resolution

735 For a discrete 2D signal of spatial size $N \times N$, the 2D discrete Fourier transform resolves frequencies
736 up to the Nyquist limit of $N/2$ cycles per image along each axis. For Tiny ImageNet ($N = 64$),
737 the maximum resolvable radial frequency is $\rho_{\max} = N/2 = 32$. Our spectral extraction pipeline
738 (Algorithm 1) partitions the frequency plane at $\rho_{\text{cut}} = \lfloor N/4 \rfloor = 16$, the half-Nyquist radius: low-
739 frequency energy E_L accumulates power at radial distance $\rho \leq 16$, and high-frequency energy E_H
740 at $\rho > 16$. This bisection ensures that both bands span comparable area in the frequency plane and
741 that $r_{H/L} = E_H/E_L$ is well-defined across the full resolvable spectrum. Because all architectures
742 process the same 64×64 input and the cutoff is a fixed fraction of spatial resolution, $r_{H/L}$ and the
743 derived $\Delta_{H/L}$ remain comparable across architectures and are not affected by aliasing beyond the
744 Nyquist limit.

745 B Training Hyperparameters

746 Table 2 lists the hyperparameter configurations for all reported experiments. Each architecture–
747 optimizer pair is trained with both BN and LN normalization variants under identical hyperparame-
748 ters; configurations marked with † exhibit block collapse (§I.1).

Algorithm 1 Spectral Bias Measurement Pipeline

Require: Trained model M with norm variants BN, LN; validation set $\mathcal{D}$; spatial cutoff $N_{\min} = 8$; bootstrap replicates $B = 10,000$

Ensure: Cross-norm spectral gap $\Delta_{H/L}$ with 95% BCa confidence interval

— **Phase 1: Spectral Extraction** —

- 1: **for** each normalization variant $\nu \in \{\text{BN}, \text{LN}\}$ **do**
- 2: Set M^ν to evaluation mode ▷ BN uses running statistics
- 3: Register forward hooks on all normalization layers
- 4: **for** each mini-batch $\mathbf{x} \in \mathcal{D}$ **do** ▷ 10 batches $\times$ 64 images
- 5: Forward pass: $M^\nu(\mathbf{x})$
- 6: **for** each hooked layer ℓ with pre/post activation $\mathbf{A}_\ell \in \mathbb{R}^{B \times C \times H \times W}$ **do**
- 7: **if** $\min(H, W) < N_{\min}$ **then**
- 8: **skip** ℓ ▷ insufficient frequency bins
- 9: **end if**
- 10: $\rho_{\text{cut}} \leftarrow \lfloor \min(H, W)/4 \rfloor$ ▷ half-Nyquist cutoff
- 11: **for** each channel $c = 1, \dots, C$ **do**
- 12: $\hat{\mathbf{F}}_c \leftarrow \text{DFT}_{2\text{D}}(\mathbf{A}_{\ell, c})$ ▷ 2D discrete Fourier transform
- 13: $P_c(u, v) \leftarrow |\hat{\mathbf{F}}_c(u, v)|^2$ ▷ power spectrum
- 14: **end for**
- 15: Radial binning: aggregate P_c into $\bar{P}(\rho)$ where $\rho = \sqrt{u^2 + v^2}$, average over B, C
- 16: $E_L \leftarrow \sum_{\rho \leq \rho_{\text{cut}}} \bar{P}(\rho), \quad E_H \leftarrow \sum_{\rho > \rho_{\text{cut}}} \bar{P}(\rho)$
- 17: $r_{H/L, \ell}^\nu \leftarrow E_H / E_L$
- 18: **end for**
- 19: **end for**
- 20: **end for**
- **Phase 2: Cross-Norm Spectral Gap** —
- 21: **for** each valid paired layer ℓ **do**
- 22: $\delta_\ell \leftarrow r_{H/L, \ell}^{\text{BN}} - r_{H/L, \ell}^{\text{LN}}$ ▷ paired difference
- 23: **end for**
- 24: $\Delta_{H/L} \leftarrow \text{median}_\ell(\delta_\ell)$ ▷ Eq. 3
- **Phase 3: Paired BCa Bootstrap** —
- 25: $\hat{\theta} \leftarrow \Delta_{H/L}$
- 26: **for** $b = 1, \dots, B$ **do**
- 27: Resample $\{\delta_\ell\}$ with replacement $\rightarrow \{\delta_\ell^*\}$
- 28: $\hat{\theta}_b^* \leftarrow \text{median}(\delta_\ell^*)$
- 29: **end for**
- 30: $z_0 \leftarrow \Phi^{-1}\left(\frac{1}{B} \sum_b \mathbf{1}[\hat{\theta}_b^* < \hat{\theta}]\right)$ ▷ bias correction
- 31: $a \leftarrow \frac{\sum_\ell (\hat{\delta}_\ell - \hat{\theta}_{(-\ell)})^3}{6 [\sum_\ell (\hat{\delta}_\ell - \hat{\theta}_{(-\ell)})^2]^{3/2}}$ ▷ acceleration via jackknife
- 32: Compute adjusted percentiles α_1, α_2 from z_0, a ▷ Efron [1987]
- 33: $\text{CI}_{95\%} \leftarrow [\hat{\theta}_{(\alpha_1)}^*, \hat{\theta}_{(\alpha_2)}^*]$
- 34: **return** $\Delta_{H/L}, \text{CI}_{95\%}$

749 **AdamW learning rate scaling.** The codebase uses a linear scaling rule: base LR $\times$ (batch_size
750 / reference_batch_size). The base LR is 10^{-3} (reference 256) for ResNet-50, DeiT-S, EfficientNet-
751 B0; 4×10^{-3} (reference 1024) for ConvNeXt-T; and 2×10^{-4} (reference 256) for SSM-based ar-
752 chitectures (VMamba-S, MambaVision-T). All AdamW configurations use $(\beta_1, \beta_2) = (0.9, 0.999)$.

753 **SGD warmup.** ResNet-50+SGD uses no warmup (BN and LN both converge without it).
754 EfficientNet-B0+SGD uses 5-epoch linear warmup; without warmup, the LN variant collapses to
755 near-chance (§I.1). All other SGD configurations use 5-epoch linear warmup.

756 **Batch size note.** Table 2 reports the *intended* batch size per configuration; actual training runs
757 deviated in two cases due to memory constraints. EfficientNet-B0+SGD used batch size 256 (vs. 128
758 for other configurations). For EfficientNet-B0+AdamW, a within-pair batch size mismatch exists:
759 the s42 pair trained BN at BS=256 and LN at BS=128, while the s123 pair trained both sides at

Table 2: Hyperparameter configurations for all reported experiments (Tiny ImageNet, 200 epochs, cosine annealing with linear warmup). All AdamW configurations use $(\beta_1, \beta_2) = (0.9, 0.999)$. AdamW LR values reflect the code’s linear scaling rule at batch size 128: base LR $\times (128 / \text{reference})$. SSM-based architectures (VMamba-S, MambaVision-T) use gradient norm clipping at 1.0.

Architecture	Optimizer	LR	WD	Mom.	Warmup	BS	Seeds
ResNet-50	SGD	0.1	5×10^{-4}	0.9	0	128	0, 7, 123
ResNet-50	AdamW	5×10^{-4}	0.05	—	5	128	42, 123
DeiT-S	AdamW	5×10^{-4}	0.05	—	5	128	42, 123, 0
DeiT-S [†]	SGD	0.1	5×10^{-4}	0.9	5	128	42
ConvNeXt-T	AdamW	5×10^{-4}	0.05	—	5	128	42, 123, 0
ConvNeXt-T [†]	SGD	0.1	5×10^{-4}	0.9	5	128	42
EfficientNet-B0	AdamW	5×10^{-4}	0.05	—	5	128	42, 123
EfficientNet-B0	SGD	0.1	5×10^{-4}	0.9	5	256	42
VMamba-S	AdamW	1×10^{-4}	0.05	—	5	128	42, 123
VMamba-S	SGD	0.1	5×10^{-4}	0.9	5	128	42
MambaVision-T	AdamW	1×10^{-4}	0.05	—	5	128	42, 123
MambaVision-T	SGD	0.1	5×10^{-4}	0.9	5	128	42

BS=256 (matched). The pooled $\hat{\theta} = -0.369$ includes both seeds; comparing the matched s123 pair alone yields $\Delta_{H/L}$ within 0.02 of the pooled value (D155). This agreement confirms that the mismatch does not affect directional conclusions.

Gradient clipping. SSM-based architectures (VMamba-S, MambaVision-T) use gradient norm clipping at 1.0; all other architectures use no gradient clipping.

B.1 Data Augmentation

Table 3 summarizes the data augmentation pipeline for each dataset, applied identically to BN and LN variants.

Table 3: Data augmentation per dataset. All datasets are normalized with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). “HFlip” denotes RandomHorizontalFlip ($p=0.5$).

Dataset	Train	Val
Tiny ImageNet (64×64)	RandomCrop(64, pad=8), HFlip	None
CIFAR-100 (32×32)	RandomCrop(32, pad=4), HFlip	None
ImageNet-100 (224×224)	RandomResizedCrop(224), HFlip	Resize(256), CenterCrop(224)

No color jitter, mixup, cutout, or label smoothing is applied. The minimal augmentation is deliberate: it avoids confounding spectral measurements with frequency-domain artifacts introduced by aggressive augmentation.

B.2 Compute Budget

All experiments were conducted on NVIDIA RTX 5090 (32 GB) GPUs. Table 4 reports approximate per-run training times for representative configurations on Tiny ImageNet (200 epochs, batch size 128, single GPU).

The registry records approximately 90 successful training runs across all architectures, datasets, and ablations, totaling an estimated 300–400 single-GPU hours. Spectral extraction (10 batches of forward passes per model) adds negligible overhead (< 1 min per run).

C Statistical Methods and Robustness Checks

Tiny ImageNet training data. The Tiny ImageNet training directory used in our experiments contained duplicate image entries due to a preprocessing artifact: each class folder included both

Table 4: Approximate per-run training time (Tiny ImageNet, 200 epochs, single GPU).

Architecture	Time (h)
ResNet-50	3–4
DeiT-Small	4–5
EfficientNet-B0	~3
ConvNeXt-Tiny	~4
VMamba-S	5–6
MambaVision-T	4–5

the original images in an `images/` subdirectory and a flat copy in the class root. PyTorch’s ImageFolder loader recursively indexes both copies; the effective training set therefore contains approximately 200,000 images (vs. the canonical 100,000). All configurations trained on the identical duplicated dataset. Because all $\Delta_{H/L}$ comparisons are *paired* (BN vs. LN under the same optimizer and data), the duplication does not affect relative spectral measurements or any quantitative claim in this paper. ImageNet-100 experiments are unaffected.

We construct 95% confidence intervals for $\Delta_{H/L}$ using the bias-corrected and accelerated (BCa) bootstrap Efron [1987] with $B = 10,000$ resamples. When BN and LN model variants share a one-to-one layer correspondence (the ℓ -th BN layer and ℓ -th LN layer occupy the same architectural position), the resampling unit is the paired per-layer difference $\delta_\ell = r_{H/L}^{\text{BN},(\ell)} - r_{H/L}^{\text{LN},(\ell)}$ within each seed; otherwise, layers are resampled independently. The point estimate is $\Delta_{H/L} = \text{median}_\ell(\delta_\ell)$, the median of the paired differences pooled across seeds. For configurations without one-to-one layer correspondence (DeiT-S), $\Delta_{H/L}$ is computed as the difference of medians: $\hat{r}_{H/L}^{\text{BN}} - \hat{r}_{H/L}^{\text{LN}}$ (independent bootstrap). Both cases report the BCa-corrected point estimate $\Delta_{H/L}$.

Anti-conservative caveat. BCa bootstrap assumes i.i.d. resampling units. Layers within a single training run share optimizer state and data order, violating independence and inflating effective sample size; reported CIs are therefore anti-conservative. As a supplementary directional consistency check, the cross-configuration sign test (§3.4) avoids the layer-independence assumption: the probability that all five AdamW configurations yield non-positive $\Delta_{H/L}$ under the null is $0.5^5 = 0.031$ (see Appendix C.3 for sensitivity to inclusion criteria); however, at $m = 5$ the test has minimal power and serves as corroborating evidence rather than primary inference. Per-configuration BCa intervals provide supplementary quantification of effect magnitude. In addition, BCa CI width scales as $1/\sqrt{n}$, so architectures with aggressive downsampling that retain few valid layers (e.g., MambaVision-T, $n \approx 6$) yield CIs roughly $3\times$ wider than ResNet-50 ($n \approx 50$). For such low- n configurations, an inconclusive result (CI spanning zero) reflects insufficient statistical power to detect an effect of moderate size, not evidence that the effect is absent.

In practice, most architectures admit pairing because BN and LN variants share the same layer structure. DeiT-S is resampled independently: our BN wrapper introduces additional normalization sites (pre-attention and post-MLP batch norms) that alter the layer indexing relative to the LN variant, breaking one-to-one correspondence across depth bins. Specifically, BN and LN variants produce different numbers of valid layers due to differing patch embedding dimensions, necessitating an independent (unpaired) bootstrap for this configuration. Both variants have identical parameter counts (21.7M); BN’s running statistics are buffers, not parameters, so the wrapper introduces no parameter-count confound. Sample sizes range from $n = 12$ to $n = 88$ (pooled across seeds) valid layers per group (Table 1). Two-sided p -values are derived from the bootstrap null distribution.

Sign-flip permutation test. As a complementary robustness check that does not rely on resampling, we implement a sign-flip permutation test (`scripts/permutation_test_theta.py`): under the null hypothesis that normalization type has no effect, the sign of each per-layer paired difference δ_ℓ is equally likely to be positive or negative. We randomly flip signs across 10,000 permutations to construct an exact null distribution for the mean paired difference $\bar{\delta}$ and derive a two-sided p -value. This test is exact under exchangeability of BN/LN labels within each layer pair and complements the BCa bootstrap (which assumes i.i.d. resampling units) and the Wilcoxon signed-rank test (which further requires symmetry of paired differences).

To control the false discovery rate across all m configurations reported in Table 1, Appendix F.1, Appendix G.2, and VMamba-S+SGD (§1.1) ($m = 13$), we apply the Benjamini–Hochberg (BH) procedure Benjamini and Hochberg [1995] at $\alpha = 0.05$; all ten significant configurations survive correction (Appendix C.1). As a supplementary robustness check, we verify significance conclusions with the Wilcoxon signed-rank test Wilcoxon [1945] on ten of the twelve configurations; all yield identical significance determinations under both methods (Appendix C.2). The Wilcoxon test assumes both independence and symmetry of the paired differences δ_ℓ ; per-layer dependence within a model and potential skewness of spectral ratio differences may violate both assumptions, so the Wilcoxon p -values should be interpreted as supplementary evidence rather than primary inference (see Appendix C.2 for discussion).

C.1 Benjamini–Hochberg Correction

Table 5 reports Benjamini–Hochberg (BH) adjusted p -values for all $m = 13$ bootstrap tests. CIFAR-100 configurations are excluded from the primary BH family because CIFAR-100’s 32×32 resolution constitutes a qualitatively different spectral regime: the Nyquist limit restricts the maximum representable spatial frequency to 16 cycles, compressing the observable frequency axis relative to Tiny ImageNet’s 32 cycles. Pooling resolution regimes into a single BH family would conflate distinct spectral phenomena (see §M.6 for sensitivity analysis). For $p < 0.001$, we use $p = 0.0005$ as a conservative upper bound. Ten configurations are significant at the uncorrected $\alpha = 0.05$ level; all ten remain significant after BH correction ($p_{\text{adj}} < 0.05$). The most marginal case is VMamba-S+SGD ($p_{\text{adj}} = 0.049$) (see Appendix E.3 for details). ConvNeXt-T+AdamW ($p_{\text{adj}} = 0.033$) is the next most marginal among the clearly significant configurations.

ConvNeXt-T outlier sensitivity. The wide bootstrap CI for ConvNeXt-T ($[-14.152, -0.121]$) is driven by high per-layer variance in $r_{H/L}$ among the 9 retained layers. IQR-based outlier detection identifies two of 27 seed-layer observations as outliers: `features.3.1.block.2` (seed 0, $\delta = -111.0$; $\text{LN } r_{H/L} = 162.3$) and `features.3.2.block.2` (seed 0, $\delta = +25.3$). Removing the top and bottom 5% of per-layer $\Delta_{H/L}$ values yields $\Delta_{H/L, \text{trim}} = -0.309$ (95% CI $[-8.871, -0.111]$, $p = 0.015$, $n = 23$). The point estimate is unchanged (the median is robust to extremes), the CI narrows by 39%, and significance strengthens ($p = 0.015$ vs. 0.028). The ConvNeXt-T result is therefore robust to outlier trimming.

Table 5: Benjamini–Hochberg adjusted p -values ($m = 13$, $\alpha = 0.05$). Configurations are sorted by raw p -value. $\Delta_{H/L}$: bootstrap median of paired BN–LN spectral differences.

Rank	Configuration	$\Delta_{H/L}$	Raw p	BH adj. p	Sig.
1	ResNet-50+SGD (TinyIN)	+0.454	<0.0001	<0.001	✓
2	DeiT-S+AdamW (TinyIN)	−0.227	< 0.0001	<0.001	✓
3	ResNet-50+SGD (IN100)	+0.357	< 0.0001	<0.001	✓
4	DeiT-S+AdamW (IN100)	−0.424	<0.001	<0.001	✓
5	EfficientNet-B0+AdamW (TinyIN)	−0.369	<0.0001	<0.001	✓
6	MambaVision-T+AdamW (TinyIN)	−35.194	0.0004	0.0009	✓
7	VMamba-S+AdamW (TinyIN)	−7.742	0.0012	0.002	✓
8	EfficientNet-B0+SGD (TinyIN)	+0.253	0.003	0.005	✓
9	ConvNeXt-T+AdamW (TinyIN)	−0.320	0.023	0.033	✓
10	VMamba-S+SGD (TinyIN)	−0.708	0.038	0.049	✓
11	DeiT-S+Lion (TinyIN)	−0.018	0.676	0.799	
12	ResNet-50+AdamW (TinyIN)	−0.016	0.802	0.869	
13	ResNet-50+AdamW (IN100)	+0.010	0.882	0.882	

C.2 Non-parametric Robustness Check

To verify that the bootstrap permutation results in Table 1 are not artifacts of the resampling procedure, we apply the Wilcoxon signed-rank test Wilcoxon [1945] as an independent non-parametric alternative. For each configuration, we compute per-layer paired differences $\delta_\ell = r_{H/L}^{\text{BN},(\ell)} - r_{H/L}^{\text{LN},(\ell)}$ at all valid layers ($\geq 8 \times 8$), pooled across seeds, and test H_0 : median $\delta = 0$ (two-sided).

Independence caveat. The per-layer Wilcoxon test treats each layer’s δ_ℓ as an independent observation. In practice, layers within the same model share training dynamics (optimizer state, data order, weight initialization), so the effective sample size is smaller than the reported n ; p -values in Table 6 may therefore be anti-conservative. We present these results as a *supplementary* robustness check; the BCa bootstrap confidence intervals in Table 1 remain the primary inferential evidence.

Symmetry assumption. The Wilcoxon signed-rank test further assumes that the distribution of paired differences δ_ℓ is symmetric around its median. Spectral ratio data may violate this assumption: architectures with heterogeneous layer geometries (e.g., ConvNeXt-T with stage-dependent downsampling, or MambaVision-T with its SSM–CNN hybrid stages) can produce skewed δ_ℓ distributions. The BCa bootstrap imposes no distributional assumptions on the test statistic (neither symmetry nor independence) and is therefore more appropriate as the primary inferential method for these data.

Table 6: Wilcoxon signed-rank test results compared with BCa bootstrap p -values from Table 1 and Appendix G.2–F.1. n : number of non-zero paired differences; W : Wilcoxon test statistic. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Configuration	n	W	Wilcoxon p	Bootstrap p
ResNet-50+SGD (TinyIN)	25	2.0	<0.001***	<0.001
ResNet-50+AdamW (TinyIN)	50	507.0	0.211	0.802
DeiT-S+AdamW (TinyIN)	50	101.0	<0.001***	<0.001
ConvNeXt-T+AdamW (TinyIN)	27	104.0	0.041*	0.023
VMamba-S+AdamW (TinyIN)	28	100.0	0.018*	0.0012
MambaVision-T+AdamW (TinyIN)	12	3.0	0.002**	0.0004
DeiT-S+Lion (TinyIN)	50	614.0	0.826	0.676
ResNet-50+AdamW (IN100)	44	492.0	0.977	0.882
ResNet-50+SGD (IN100)	44	5.0	<0.001***	<0.001
DeiT-S+AdamW (IN100)	25	0.0	<0.001***	<0.001

Sample sizes differ from Table 1 because the Wilcoxon test requires layer-level pairing: for multi-seed configurations, per-layer $\Delta_{H/L}$ values are averaged across seeds before pairing ($n = 25$ layers for ResNet-50), whereas the bootstrap resamples seed–layer combinations independently ($n = 50$). For DeiT-S+AdamW (single seed), $n = 25$ seed–layer combinations. Two of the three ImageNet-100 configurations (ResNet-50+AdamW, DeiT-S+AdamW) are single-seed; for these, n equals the number of valid layers per model without cross-seed averaging. ResNet-50+SGD (IN100) has two seeds (s42, s123), with $n = 44$ per seed.

All ten configurations in Table 6 yield identical significance conclusions under the Wilcoxon test and the BCa bootstrap: seven are significant at $\alpha = 0.05$ under both methods, and three are non-significant under both. The exact p -values differ (the Wilcoxon test is slightly more conservative for VMamba-S, 0.018 vs. 0.0012, and slightly less conservative for ConvNeXt-T, 0.041 vs. 0.023), but no configuration changes significance status. Note that ConvNeXt-T is a borderline case: the Wilcoxon $p = 0.041$ is close to the $\alpha = 0.05$ threshold (the corresponding bootstrap $p = 0.023$, BH-adjusted to 0.033, is similarly borderline), and the non-independence of layers (see caveat above) means the true p -value may be higher. This concordance confirms that the reported findings are robust to the choice of statistical test. The BCa bootstrap (Table 1) remains the primary inferential method because it imposes neither the symmetry assumption of the Wilcoxon test nor the independence assumption violated by within-model layer correlations; the Wilcoxon results serve as supplementary corroboration.

Cross-dataset replication. Both key configurations replicate their $\Delta_{H/L}$ sign across Tiny ImageNet and ImageNet-100: ResNet-50+SGD yields +0.454 (TinyIN) and +0.430/ + 0.266 across two IN100 seeds; DeiT-S+AdamW yields −0.227 (TinyIN) and −0.424 (IN100). This cross-dataset consistency provides evidence that the directional findings are not dataset-specific artifacts, independent of any distributional assumption required by either the Wilcoxon or bootstrap tests.

Per-seed median aggregation. As an additional check, we reduce each seed to a single summary statistic, the median δ_ℓ across layers, and apply the Wilcoxon test on n_{seeds} values (typically 2–3). This per-seed aggregation is conservative but statistically less powerful: it respects the non-

independence of layers within a model (each seed contributes exactly one observation), but with only 2–3 seeds per configuration the test has very low power and is expected to be non-significant even for genuine effects. The purpose is to confirm that our conclusions do not depend on the inflated n from pooling non-independent layers, not to serve as a standalone significance test. Results are available via `scripts/wilcoxon_test.py -mode per-seed`.

Per-seed direction consistency. For configurations trained with multiple random seeds, we verify that the sign of per-seed $\Delta_{H/L}$ agrees across seeds; this agreement confirms that pooled results are not driven by a single anomalous seed. Table 7 summarizes the per-seed median $\Delta_{H/L}$ for all multi-seed configurations.

Table 7: Per-seed median $\Delta_{H/L}$ direction. For each seed, $\Delta_{H/L}$ is the median of per-layer paired differences across valid layers ($\geq 8 \times 8$). “+” / “−” / “ ≈ 0 ” indicate the sign of the per-seed estimate. Direction unanimity holds for all configurations with significant pooled results.⁸

Configuration	Seed 42	Seed 123	Seed 0	Seed 7	Pooled $\Delta_{H/L}$
ResNet-50+SGD (TinyIN)	—	+	+	+	+0.454***
ResNet-50+AdamW (TinyIN)	≈ 0	−	−	−	−0.016
DeiT-S+AdamW (TinyIN)	−	−	−	−	−0.227***
ConvNeXt-T+AdamW (TinyIN)	−	−	−	−	−0.320*
VMamba-S+AdamW (TinyIN)	−	−	−	−	−7.742***
MambaVision-T+AdamW (TinyIN)	−	−	−	−	−35.194***
DeiT-S+Lion (TinyIN)	−	+	−	−	−0.018

All five significant configurations (ResNet-50+SGD, DeiT-S+AdamW, ConvNeXt-T+AdamW, VMamba-S+AdamW, MambaVision-T+AdamW) show unanimous per-seed sign agreement; the pooled direction is not an artifact of seed averaging. The two non-significant configurations exhibit mixed or near-zero per-seed signs, consistent with their null pooled results. For the two configurations with ImageNet-100 replication, the IN100 $\Delta_{H/L}$ signs are consistent with all TinyIN seeds: ResNet-50+SGD shows +0.430 (s42) and +0.266 (s123), both positive; DeiT-S+AdamW shows −0.424 (s42). These cross-dataset results corroborate the per-seed direction pattern. For DeiT-S+Lion, the per-seed estimates ($\Delta_{H/L} = -0.034$ at seed 42, +0.015 at seed 123) have opposing signs with non-significant p -values (0.32 and 0.53 respectively; Appendix G.2). This pattern confirms a genuine absence of systematic spectral bias rather than cancellation of opposing seed-level effects.

C.3 Sign Test Sensitivity

The cross-configuration sign test (§3.4) counts independent configurations yielding non-positive $\Delta_{H/L}$. Table 8 reports the sign test under four inclusion criteria, ranging from a conservative $m = 4$ (excluding the ρ_{cut} -sensitive MambaVision-T) to an expansive $m = 7$.

Table 8: Sign test sensitivity to inclusion criteria. All configurations listed yield non-positive $\Delta_{H/L}$. **Bold:** primary count used in the main text.

m	$p = 0.5^m$	Configurations included
4	0.0625	Conservative: DeiT-S, ConvNeXt-T, VMamba-S, EfficientNet-B0 (excluding ρ_{cut} -sensitive MambaVision-T)
5	0.031	$m=4$ + MambaVision-T (all AdamW, Tiny ImageNet)
6	0.016	$m=5$ + cross-dataset: DeiT-S+AdamW (ImageNet-100, 224×224)
7	0.008	$m=6$ + VMamba-S+SGD ($\Delta_{H/L} = -0.708$)

⁸ResNet-50+AdamW (IN100) and DeiT-S+AdamW (IN100) are excluded from per-seed analysis as each was trained with a single seed. ResNet-50+SGD (IN100) has two seeds; per-seed results are shown in Table 10.

Inclusion criteria. The inclusive count ($m = 6$) comprises five same-dataset configurations (each architecture contributing one AdamW observation on Tiny ImageNet) plus DeiT-S+AdamW on ImageNet-100 ($\Delta_{H/L} = -0.424$) as a cross-dataset replication. Because ImageNet-100 uses a different resolution (224×224 vs. 64×64) and different training images, it constitutes a partially independent observation. We exclude ResNet-50+AdamW because it serves as the LINCHPIN ablation control (§4.2): its near-zero $\Delta_{H/L}$ (-0.016 , $p = 0.802$) is the expected outcome under the null hypothesis that SGD, not normalization, drives the spectral gap, and including it would conflate the test variable (normalization effect across architectures) with the control variable (optimizer ablation on a single architecture).

The primary count ($m = 5$, $p = 0.031$) includes all five same-dataset AdamW configurations, avoiding double-counting any architecture across datasets. The conservative $m = 4$ excludes MambaVision-T, whose $\Delta_{H/L}$ changes sign across ρ_{cut} values (§4.1; Appendix K.1). At $m = 7$, VMamba-S+SGD ($\Delta_{H/L} = -0.708$, $p = 0.038$) is further included as a non-AdamW configuration, broadening the test from “all AdamW configurations are non-positive” to “all non-SGD-CNN configurations are non-positive.”

The directional conclusion (non-positive $\Delta_{H/L}$ under AdamW) shows consistent direction across all four inclusion criteria, though the most conservative count ($m = 4$, excluding MambaVision-T) yields $p = 0.0625$, which does not reach the conventional $\alpha = 0.05$ threshold.

939 C.4 ConvNeXt-T Trimmed Bootstrap Analysis

ConvNeXt-T retains only 9 of 23 normalization layers above the 8×8 spatial resolution threshold; the resulting sample comprises $n = 27$ seed-layer observations across three seeds. The bootstrap CI $[-14.152, -0.121]$ is wide relative to the point estimate ($\Delta_{H/L} = -0.320$), which raises the concern that outlier layers may dominate the CI.

Outlier identification. IQR-based detection identifies two of 27 observations as outliers: `features.3.1.block.2` (seed 0, $\delta = -111.0$; LN $r_{H/L} = 162.3$) and `features.3.2.block.2` (seed 0, $\delta = +25.3$). Both outliers reside in stage 3, the final retained stage before the resolution filter excludes deeper layers.

Trimmed bootstrap. Removing the top and bottom 5% of per-layer δ_ℓ values and recomputing the BCa bootstrap ($B = 10,000$) yields:

- Point estimate: $\Delta_{H/L, \text{trim}} = -0.309$ (unchanged; the median is robust to extremes)
- 95% CI: $[-8.871, -0.111]$ (39% narrower than untrimmed $[-14.152, -0.121]$)
- $p = 0.015$ (strengthened from $p = 0.023$)
- $n = 23$ (after removing 4 of 27 observations)

The sign of $\Delta_{H/L}$ is stable under trimming: the negative direction is preserved with a narrower CI and stronger significance. The identical point estimate before and after trimming demonstrates that the median-based estimator is not driven by the outlier layers.

Implication. ConvNeXt-T’s negative $\Delta_{H/L}$ under AdamW is robust to outlier trimming. The wide untrimmed CI reflects genuine per-layer variance in stage 3 rather than a fragile result driven by a single anomalous observation.

960 D Normalized Effect Sizes and Per-Seed Measurements

Baseline $r_{H/L}$ spans two orders of magnitude across architectures, so raw $\Delta_{H/L}$ is not directly comparable. We define a midpoint-normalized variant:

$$\Delta_{H/L}^{\text{rel}} = \frac{\Delta_{H/L}}{(\tilde{r}_{H/L}^{\text{BN}} + \tilde{r}_{H/L}^{\text{LN}}) / 2} \times 100\%. \quad (4)$$

This maps raw gaps (e.g., -0.227 for DeiT-S vs. -35.194 for MambaVision-T) to comparable relative magnitudes (-37.9% vs. -34.9%).

The approximately two-order-of-magnitude variation in $|\Delta_{H/L}|$ across architectures (Table 1; §5) largely reflects differences in baseline spectral ratio rather than in normalization sensitivity. Table 9

967 reports effect sizes normalized by the BN-side median post-normalization $\tilde{r}_{H/L}^{\text{BN}}$ for each configura-
 968 tion, removing the architecture-dependent scale factor.

Table 9: Normalized effect sizes ($\Delta_{H/L} / \tilde{r}_{H/L}^{\text{BN}}$). $\tilde{r}_{H/L}^{\text{BN}}$: BN-side median post-normalization $r_{H/L}$, pooled across seeds and valid layers ($\geq 8 \times 8$). After normalization, the sign pattern (positive under SGD, non-positive under AdamW) is preserved, and the raw two-order-of-magnitude spread in $|\Delta_{H/L}|$ compresses substantially. Values marked † are approximate (sourced from preliminary per-seed medians).

Configuration	$\Delta_{H/L}$	$\tilde{r}_{H/L}^{\text{BN}}$	$\Delta_{H/L} / \tilde{r}_{H/L}^{\text{BN}}$
ResNet-50+SGD (TinyIN)	+0.454	0.83 †	+0.58 †
ResNet-50+SGD (IN100)	+0.357	0.657	+0.543
ResNet-50+AdamW (TinyIN)	−0.016	0.669	−0.024
ResNet-50+AdamW (IN100)	+0.010	0.848	+0.012
DeiT-S+AdamW (TinyIN)	−0.227	0.52 †	−0.44 †
DeiT-S+AdamW (IN100)	−0.424	1.018	−0.417
DeiT-S+Lion (TinyIN)	−0.018	0.929	−0.019
ConvNeXt-T+AdamW (TinyIN)	−0.320	16.165	−0.020
VMamba-S+AdamW (TinyIN)	−7.742	56.9 †	−0.14 †
MambaVision-T+AdamW (TinyIN)	−35.194	68.8	−0.51

969 Across all ten configurations, the normalized effects range from −0.42 (DeiT-S+AdamW, TinyIN)
 970 to +0.618 (ResNet-50+SGD on IN100), all within the same order of magnitude, whereas the raw
 971 $\Delta_{H/L}$ values span from −35.194 to +0.454 ($\sim 78\times$). The sign dichotomy (positive only under
 972 SGD; non-positive under AdamW/Lion) is invariant to this rescaling. This invariance confirms that
 973 the directional finding is not an artifact of architecture-dependent magnitude differences in $r_{H/L}$.

974 D.1 Per-Seed Spectral Bias Measurements

975 Table 10 reports $\Delta_{H/L}$ for each seed–configuration pair where per-seed bootstrap results are avail-
 976 able, and pooled $\Delta_{H/L}$ (across seeds) for the remaining multi-seed configurations. Per-seed $\Delta_{H/L}$
 977 is the BCa-corrected median of per-layer paired differences δ_ℓ within a single seed; pooled $\Delta_{H/L}$
 978 resamples across all seeds jointly (as in Table 1). For all five significant multi-seed configurations,
 979 every individual seed agrees in sign with the pooled estimate (Table 7); the two non-significant con-
 980 figurations (ResNet-50+AdamW, DeiT-S+Lion) exhibit mixed per-seed signs, consistent with a null
 981 pooled result.

982 D.1.1 ResNet-50+SGD Per-Seed Detail

983 Table 11 reports per-seed validation accuracy and paired spectral gap for the ResNet-50+SGD con-
 984 figuration on Tiny ImageNet. BN converges across all four seeds ($\sigma \approx 0.4$ pp); LN seed 42 (v3)
 985 collapses at epoch 52 (Appendix H) and is excluded from all spectral analyses. The three surviving
 986 LN seeds yield per-seed $\Delta_{H/L}$ of +0.454 (s0), +0.498 (s123), and +0.335 (s7).

987 **Pooling protocol.** BN trains stably across all four seeds, but the paired analysis requires matching
 988 BN–LN seed pairs. Because LN seed 42 collapsed, three seed pairs (s0, s123, s7) contribute to the
 989 pooled bootstrap ($n = 25 \times 3 = 75$ layer pairs). BN seed 42 is not used in the primary paired
 990 analysis but is included in the BN leave-one-out sensitivity check (Appendix K.4).

991 E Architecture-Specific Analyses

992 E.1 EfficientNet-B0 Details

993 EfficientNet-B0 Tan and Le [2019] is included in the main comparison (Table 1) with AdamW,
 994 contributing a sixth architecture to the cross-norm spectral analysis. Its depthwise-separable con-
 995 volution blocks produce 49 normalization layers, of which 16 meet the $\geq 8 \times 8$ spatial resolution
 996 threshold for spectral extraction. The paired bootstrap (two seeds pooled) yields $\Delta_{H/L} = -0.369$

Table 10: Per-seed spectral bias measurements across configurations. $\Delta_{H/L}$: median cross-norm spectral gap (BN–LN). n : number of valid layers ($\geq 8 \times 8$). Rows labelled with multiple seeds (e.g., “s42+s123”) report the pooled bootstrap estimate; single-seed rows report the per-seed bootstrap. “Running” denotes experiments in progress at time of writing. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Configuration	Seed	$\Delta_{H/L}$	Sign	p	n
<i>ResNet-50</i>					
+SGD (TinyIN)	s0	+0.454	+	<0.0001***	25
	s123	+0.498	+	<0.0001***	25
	s7	+0.335	+	<0.0001***	25
	s42	crashed at epoch 52 (Appendix H)			
+AdamW (TinyIN)	Pooled (s0+s123+s7)	+0.454	+	< 0.0001***	75
	s42+s123	−0.016	≈ 0	0.802	50
	s42	+0.024	+	0.842	25
	s123	−0.031	−	0.542	25
+SGD (IN100)	s42+s123	+0.357	+	< 0.0001***	88
	s42	+0.430	+	<0.001***	44
	s123	+0.266	+	<0.001***	44
+AdamW (IN100)	s42	+0.010	+	0.882	44
<i>DeiT-S</i>					
+AdamW (TinyIN)	s0	−0.227	−	< 0.0001***	25
+Lion (TinyIN)	s42	−0.034	−	0.32	25
	s123	+0.015	+	0.53	25
+AdamW (IN100)	s42	−0.424	−	<0.001***	25
<i>ConvNeXt-T</i>					
+AdamW (TinyIN)	s42+s123+s0	−0.320	−	0.023*	27
<i>VMamba-S</i>					
+AdamW (TinyIN)	s42+s123	−7.742	−	0.0012**	28
+SGD (TinyIN)	s42	−0.708	−	0.038 ^(†)	14
<i>MambaVision-T</i>					
+AdamW (TinyIN)	s42+s123	−35.194	−	0.0004***	12
+SGD (TinyIN)	s42	+15.533	+	0.203	6
<i>EfficientNet-B0</i>					
+AdamW (TinyIN)	s42+s123	−0.369	−	<0.0001***	32
+SGD (TinyIN)	s42	+0.253	+	0.003**	16

Table 11: ResNet-50+SGD per-seed detail (Tiny ImageNet, 200 epochs). Per-seed $\Delta_{H/L}$: BCa-corrected median of $n=25$ per-layer BN–LN paired differences within each seed pair. Pooled $\Delta_{H/L}$ resamples across all three surviving LN seeds ($n=75$). exp_ids: res_resnet50_sgd_{bn,ln}_{s0,s42,s123,s7}.

Seed pair	BN val (%)	LN val (%)	$\Delta_{H/L}$	p	Note
s0	58.91	43.31	+0.454	<0.0001	
s123	58.75	43.79	+0.498	<0.0001	
s7	58.86	42.94	+0.335	<0.0001	
s42	57.97	—	—	—	LN crashed ep52
Pooled (s0+s123+s7)	—	—	+0.454	< 0.0001	$n=75$, CI [+0.270, +0.498]

997 with 95% BCa CI $[-1.332, -0.227]$ ($p < 0.0001$, $n = 32$). This non-positive $\Delta_{H/L}$ is consistent
998 with the AdamW pattern across all other architectures.

999 **SGD+LN collapse.** Without learning rate warmup, EfficientNet-B0+SGD+LN exhibits represen-
1000 tational collapse identical to the DeiT-S and ConvNeXt-T SGD failures (§I.1), with validation
1001 accuracy at near-chance ($\sim 2\%$). Adding 5-epoch linear warmup restores convergence (val. ac-
1002 curacy $\sim 46\%$) and allows paired spectral extraction: $\Delta_{H/L} = +0.253$ ($p = 0.003$, 95% CI
1003 $[+0.040, +0.424]$, $n = 16$). This result confirms the SGD positive bias on a second CNN architec-

1004 ture. The SGD+BN variant trains normally (53.57% val accuracy); the collapse without warmup is
1005 therefore normalization-specific rather than architecture-wide.

1006 **LINCHPIN replication.** Under AdamW, EfficientNet-B0 exhibits the same spectral neutraliza-
1007 tion as ResNet-50: BN yields median $r_{H/L} = 0.664$ while LN yields 1.099. LN therefore pre-
1008 serves slightly more high-frequency energy, opposite to the SGD direction. This result replicates
1009 the LINCHPIN pattern on a second CNN architecture ($n_{\text{arch}} = 2$), strengthening the argument that
1010 optimizer regime shapes normalization-induced spectral bias.

1011 E.2 MambaVision-T+SGD Spectral Analysis

1012 MambaVision-T under SGD yields a positive point estimate $\Delta_{H/L} = +15.533$ with 95% BCa CI
1013 $[-2.080, +40.201]$ ($p = 0.203$, $n = 6$ paired layers, single seed s42). The direction aligns with
1014 ResNet-50+SGD ($\Delta_{H/L} = +0.454$) rather than VMamba-S+SGD ($\Delta_{H/L} = -0.708$), but the
1015 confidence interval spans zero and the result does not reach statistical significance.

1016 The low statistical power stems from MambaVision-T’s aggressive spatial downsampling: only 6
1017 of its normalization layers produce feature maps at or above the 8×8 spatial resolution threshold
1018 required for reliable spectral extraction. At $n = 6$, the BCa CI is approximately $3.6 \times$ wider than for
1019 ResNet-50 ($n = 25$ per seed), consistent with the $1/\sqrt{n}$ scaling noted in Appendix C.

1020 **Extraction-method correction.** An earlier version of the spectral extraction pipeline inadver-
1021 tently included AdamW runs in the MambaVision-T+SGD sample pool; the resulting point estimate
1022 was near zero ($\Delta_{H/L} = -0.013$). After restricting the extraction to the six pure SGD layers (seed 42
1023 only), $\Delta_{H/L}$ reversed sign to $+15.533$, a direction consistent with the SGD positive-bias pattern ob-
1024 served in ResNet-50. The corrected result remains statistically non-significant ($p = 0.203$, 95%
1025 BCa CI $[-2.080, +40.201]$), as the confidence interval spans zero. With only $n = 6$ observations
1026 from a single seed, the sample is too small for reliable inference; we report this configuration as a
1027 supplementary observation outside the main BH-corrected framework ($m = 13$, Table 5).

1028 We therefore classify this result as *inconclusive*: the positive direction is suggestive but not con-
1029 firmatory. Multi-seed replication would be needed to determine whether MambaVision-T’s hybrid
1030 CNN-SSM architecture preserves the CNN-like positive spectral bias under SGD or whether the
1031 single-seed positive estimate is a sampling artifact.

1032 E.3 VMamba-S+SGD Spectral Analysis

1033 VMamba-S under SGD yields $\Delta_{H/L} = -0.708$ with 95% BCa CI $[-14.753, 0.000]$ ($p = 0.038$,
1034 $n = 14$ paired layers, seed s42). The result is significant after BH correction ($p_{\text{adj}} = 0.049$, rank 10
1035 of $m = 13$; Table 5).

1036 **Block collapse and low statistical power.** VMamba-S retains only 14 of 50 normalization layers
1037 above the 8×8 spatial resolution threshold. Furthermore, VMamba-S exhibits partial block collapse
1038 under SGD: BN and LN representations converge in deeper layers, inflating per-layer $\Delta_{H/L}$ vari-
1039 ance. The ~ 14 pp BN-LN accuracy gap under SGD (50.75% vs. 36.70%) further confounds causal
1040 interpretation of the spectral measurement (§I.1).

1041 VMamba-S+SGD exhibits $\Delta_{H/L} = -0.708$ (BH-adjusted $p = 0.049$), significant after BH correc-
1042 tion. The raw p -value (0.038) and adjusted p -value both fall below $\alpha = 0.05$, though the 95% CI
1043 $[-14.753, 0.000]$ spans zero and the 14 pp accuracy confound warrants cautious interpretation.

1044 **Data-contamination correction.** An earlier version of the VMamba-S+SGD analysis pooled
1045 seeds 42 and 123; the pooled estimate was $\Delta_{H/L} = -6.861$. This estimate was an artifact of
1046 NaN contamination: the seed-123 LN variant diverged to NaN from step 1 and never produced a
1047 valid training run. Including the corrupted seed inflated the pooled median difference by an order
1048 of magnitude. After restricting the analysis to seed 42 only (whose training data are clean), $\Delta_{H/L}$
1049 corrects to -0.708 (raw $p = 0.038$, BH-adjusted $p = 0.049$, $n = 14$). All VMamba-S+SGD values
1050 reported in Table 1 and throughout the paper reflect the corrected seed-42-only estimate.

F Cross-Dataset Validation

F.1 ImageNet-100 Scale-Up Details

The preceding results are obtained on Tiny ImageNet (64×64 , 200 classes). To test whether the SGD-induced positive spectral gap is an artifact of the low-resolution training regime, we replicate the ResNet-50 BN vs. LN comparison under SGD on ImageNet-100 (100 classes, $\sim 64K$ training images, 224×224).

The positive spectral gap reproduces across two seeds: s42 yields $\Delta_{H/L} = +0.430$, 95% CI $[+0.354, +0.603]$, $p < 0.001$; s123 yields $\Delta_{H/L} = +0.266$, 95% CI $[+0.190, +0.349]$ ($n = 44$ per seed). Both directions are consistent with the Tiny ImageNet result ($\Delta_{H/L} = +0.454$), and both CIs exclude zero. The inter-seed variability is notably higher on IN100 ($+0.430$ vs. $+0.266$, a $\sim 38\%$ relative difference) than on Tiny ImageNet ($+0.454$ vs. $+0.498$, $< 10\%$), likely reflecting the smaller per-seed sample size ($n = 44$ vs. 25 on TinyIN); crucially, both seeds agree in sign and significance.

The optimizer contrast also holds at this scale: ResNet-50+AdamW on the same ImageNet-100 data yields $\Delta_{H/L} = +0.010$ ($p = 0.882$, $n = 44$), indistinguishable from zero. The point estimate decreases from $+0.454$ (Tiny ImageNet) to $+0.357$ (ImageNet-100). This attenuation may reflect richer frequency content at 224×224 resolution.

Validation accuracy follows the expected pattern: BN (78.68%) substantially outperforms LN (62.78%) under SGD, a gap of ~ 16 percentage points consistent with BN’s advantage in the SGD regime on Tiny ImageNet.

DeiT-S+AdamW on ImageNet-100. To test whether the DeiT-S negative spectral gap generalizes beyond 64×64 , we train DeiT-S with BN and LN under AdamW on ImageNet-100 (224×224 , 100 classes) for 200 epochs (seed 42). The BN variant reaches 46.62% validation accuracy; the LN variant achieves 44.22%. The spectral gap is $\Delta_{H/L} = -0.424$, 95% CI $[-0.559, -0.391]$, $p < 0.001$, $n = 25$. The direction is preserved at higher resolution. The key finding is directional consistency: DeiT-S’s negative spectral bias under AdamW is not an artifact of the 64×64 training regime.

These results extend the scale-up validation from a single architecture to two architectures spanning the CNN–Transformer divide, mitigating the “single dataset” threat. Note that most ImageNet-100 experiments use a single seed ($n = 25$ layers per configuration); statistical power is correspondingly lower. The exception is ResNet-50+SGD, which was replicated with two seeds (s42, s123; $n = 88$ pooled).

F.2 CIFAR-100 Scale-Up Details

To test whether the optimizer-driven spectral bias generalizes to a lower-resolution, coarser-grained regime, we train ResNet-50 under the full 2×2 factorial (BN/LN $\times$ SGD/AdamW) on CIFAR-100 (100 classes, 32×32 , 50K training images) for 200 epochs with batch size 256, seed 42.

Training setup. We adapt hyperparameters from Table 2 with batch size 256 (vs. 128 for Tiny ImageNet). SGD configurations use $\text{lr} = 0.1$, momentum 0.9, weight decay 5×10^{-4} , cosine schedule. AdamW configurations use $\text{lr} = 10^{-3}$, $(\beta_1, \beta_2) = (0.9, 0.999)$, weight decay 0.05, cosine schedule.

Validation accuracy. Table 12 summarizes the training outcomes. All four configurations reach near-perfect training accuracy ($\geq 99.9\%$), but validation performance diverges:

AdamW+LN overfitting analysis. The AdamW+LN configuration exhibits the most severe overfitting in our study: training accuracy exceeds 99.9% while validation peaks at 67.9%, a 32.1 pp gap that exceeds all other CIFAR-100 configurations and the analogous Tiny ImageNet setup. Two factors compound to produce this effect. First, CIFAR-100’s 32×32 resolution and 50K-sample training set provide limited data diversity, facilitating memorization. Second, LN computes moments per-sample and therefore lacks the implicit mini-batch regularization that BN provides via batch-dependent statistics Ioffe and Szegedy [2015], Luo et al. [2019]; this regularization deficit

Table 12: CIFAR-100 ResNet-50 training outcomes (seed 42; SGD+LN uses a 5-epoch linear warmup).

Configuration	Train acc	Val acc	Gap (pp)	Epochs
SGD+BN	100.0%	77.6%	22.4	200
SGD+LN	100.0%	76.1%	23.9	200
AdamW+BN	100.0%	75.7%	24.3	200
AdamW+LN	100.0%	67.9%	32.1	200

Table 13: CIFAR-100 paired spectral bias $\Delta_{H/L}$ (BN–LN). Reported separately from Table 1 due to qualitatively different frequency regime at 32×32 resolution (see §5).

Configuration	$\Delta_{H/L}$	95% CI	p	n
ResNet-50+SGD (C100)	+0.032	$[-0.085, +0.103]$	0.879	44
ResNet-50+AdamW (C100)	+0.093	$[+0.038, +0.159]$	<0.0001	44

leaves the network more susceptible to overfitting, particularly under adaptive optimizers that can exploit per-parameter curvature to memorize training examples. The train–val gap grows monotonically along both axes: replacing BN with LN adds ~ 1.5 – 7.8 pp, and replacing SGD with AdamW adds ~ 1.9 – 8.2 pp. This pattern confirms a factorial interaction between normalization type, optimizer, and generalization (see also §5).

Impact on spectral measurement. Overfitting does not compromise the paired $\Delta_{H/L}$ inference for three reasons. (a) Spectral features are extracted from the best-validation checkpoint (the epoch with highest validation accuracy), not the final epoch; this ensures that the spectral measurement reflects a model that has not yet entered the pure-memorization regime. (b) The accuracy-confound falsification established by the LINCIPIN experiment (§5) demonstrates that accuracy gap magnitude does not predict $\Delta_{H/L}$ sign, the same dissociation that motivated the CIFAR-100 extension. (c) The paired comparison $\Delta_{H/L} = r_{H/L}^{\text{BN}} - r_{H/L}^{\text{LN}}$ eliminates absolute accuracy as a confound: both BN and LN models are evaluated at their respective best checkpoints, and the *sign* of the spectral difference is the quantity of interest. We note that these results are preliminary (single seed per configuration) and await multi-seed replication to assess variance.

G Optimizer Ablations

G.1 Lion Optimizer Configuration

This subsection details the Lion optimizer experiments, including the ResNet-50 training failure (§G.1.1) and the DeiT-S pooling protocol (§G.1.2).

G.1.1 ResNet-50+Lion: Training Failure

We attempted Lion Chen et al. [2023] on ResNet-50 using the default hyperparameters recommended for ViTs ($\lambda = 1.0$, $\text{lr} = 10^{-4}$, $\beta = (0.9, 0.99)$). Both normalization variants failed:

- **BN:** Numerical instability manifesting as NaN validation loss in 46% of epochs, with onset at approximately epoch 40. Training never recovers; the NaN rate increases monotonically after onset.
- **LN:** Representational collapse to near-chance accuracy ($\text{val_acc} = 2.93\%$ at epoch 155, where chance is 0.5% for 200 classes). Unlike the BN failure, no NaN events occur; the network trains but learns a degenerate representation.

We attribute both failures to the excessively high weight decay $\lambda = 1.0$, which Chen et al. Chen et al. [2023] recommend for ViTs but which induces catastrophic regularization on ResNet-50’s convolutional parameters. The BN failure likely reflects parameter magnitudes being driven below numerical stability thresholds by aggressive decay, while the LN failure reflects a slower representational starvation as useful features are progressively penalized. Seed 123 experiments were cancelled after seed 42 exhibited identical failure modes under both normalizations.

1134 **Impact on main conclusions.** The ResNet-50+Lion failure does not affect any quantitative claim
 1135 in the main text. All reported $\Delta_{H/L}$ values (Table 1) derive from SGD and AdamW experiments,
 1136 which trained successfully. The failure serves as qualitative evidence that optimizer–architecture
 1137 compatibility is a non-trivial constraint (Appendix G.2).

1138 G.1.2 DeiT-S+Lion: Pooling Protocol

1139 Adopting architecture-appropriate hyperparameters ($\lambda = 0.3$, $\text{lr} = 3 \times 10^{-5}$) for DeiT-S resolved
 1140 all instability. Both normalizations trained stably across all seeds. The DeiT-S Lion $\Delta_{H/L}$ analysis
 1141 pools seeds 42 and 123 ($n = 2$); a third seed (seed 0) was trained with batch size 64 rather than 128
 1142 due to a configuration error, and is excluded from the pooled bootstrap because batch normalization
 1143 statistics are directly affected by batch size (D058).

1144 G.2 Lion Optimizer Full Analysis

1145 As a directional probe beyond the SGD–AdamW axis, we introduce Lion Chen et al. [2023] as a third
 1146 optimizer. Lion uses sign-based updates with decoupled weight decay, producing a qualitatively
 1147 different optimization trajectory from both SGD (momentum-only) and AdamW (adaptive second
 1148 moment).

1149 **Weight decay confound.** Chen et al. [2023] recommend $\lambda = 0.3$ for Lion on Vision
 1150 Transformers, compared to $\lambda = 0.05$ used for AdamW in our DeiT-S experiments. Because the
 1151 update-rule change and the $6\times$ increase in weight decay are introduced simultaneously, the Lion
 1152 results cannot disentangle optimizer dynamics from regularization strength; we present them as a
 1153 directional complement, not an isolated optimizer effect.

1154 **Experimental design.** We adopt the Chen et al. recommended hyperparameters ($\text{lr} = 3 \times 10^{-5}$,
 1155 $\lambda = 0.3$, $\beta = (0.9, 0.99)$) and train DeiT-S under both BN and LN with seeds 42 and 123. ResNet-
 1156 50 under Lion fails to train under either normalization (see §G.1.1).

1157 **DeiT-S+Lion: stable training.** DeiT-S under Lion trains stably under both BN and LN, with
 1158 zero NaN events across all runs. The BN variant achieves 41.08% validation accuracy (seed 42);
 1159 the BN–LN accuracy gap is negligible (~ 0.2 pp). The ~ 7 pp accuracy gain over DeiT-S+AdamW
 1160 likely reflects the stronger regularization imposed by $\lambda = 0.3$.

1161 **Spectral gap.** Per-seed analysis yields $\Delta_{H/L} = -0.034$ ($p = 0.32$, seed 42) and $\Delta_{H/L} = +0.015$
 1162 ($p = 0.53$, seed 123); neither estimate is significant, and their signs are opposing. Pooling across
 1163 seeds: $\Delta_{H/L} = -0.018$ (95% BCa CI $[-0.092, +0.042]$, $p = 0.676$; $n = 50$ layers pooled across
 1164 2 seeds).

1165 The CI spans zero; Lion neutralizes the BN–LN spectral gap on DeiT-S more thoroughly than
 1166 AdamW does on the same architecture: $|\Delta_{H/L}| = -0.018$ under Lion versus $\Delta_{H/L} = -0.227$
 1167 ($p < 0.0001$) under AdamW. On DeiT-S, the three optimizers yield qualitatively distinct spectral
 1168 regimes: SGD induces block collapse, AdamW produces a significant negative $\Delta_{H/L}$, and Lion
 1169 produces a genuine null. The 95% CI $[-0.089, +0.045]$ excludes effects of practical relevance; the
 1170 null result reflects a genuine absence of systematic spectral bias rather than inadequate sensitivity.

1171 **Implications.** Because Lion coverage is limited to a single architecture ($n = 1$) with a confounded
 1172 weight decay change, these results are preliminary. The three-optimizer matrix for ResNet-50 (SGD:
 1173 positive; AdamW: zero; Lion: training failure) and DeiT-S (SGD: collapse; AdamW: negative; Lion:
 1174 null) demonstrates that optimizer–architecture compatibility is a consistently observed correlate of
 1175 spectral behavior.

1176 G.3 Constant-LR Mechanism Analysis

1177 To determine whether the spectral reshaping is a property of the SGD–BN interaction or a byproduct
 1178 of cosine LR annealing, we trained ResNet-50 with BN and LN under SGD with a *constant* learning
 1179 rate of 0.1 for 200 epochs across two seeds.

Under constant LR, BN converges consistently: s42 reaches 37.82% best validation accuracy (epoch 165) and s123 reaches 38.73% (epoch 65). The LN variant fails in both seeds: s123 collapses to chance (0.50%) after initially reaching 22.24% (epoch 54), while s42 plateaus at $\sim 14\%$.

Spectral trajectory. The epoch-resolved $\Delta_{H/L}$ under constant LR (Figure 3) is positive at all 20 spectral checkpoints (epochs 10–200), with median values ranging from $+0.48$ to $+0.70$. The constant-LR trajectory initially exceeds the cosine baseline, peaking at $\sim +0.70$ near epoch 50 versus $\sim +0.59$ under cosine. This elevation is consistent with a sustained high learning rate prolonging spectral divergence. After the peak, the trajectory fluctuates within $[+0.51, +0.66]$ through epochs 100–200 rather than decaying. This stability confirms that the late-training decline under cosine tracks the LR decay rather than an intrinsic saturation.

The constant-LR $\Delta_{H/L}$ at the s42 best checkpoint (epoch 165): $\Delta_{H/L} = +0.5684$, $p < 0.0001$ (95% BCa CI: $[+0.241, +0.750]$, $n = 25$ layers). At the training endpoint (epoch 200): $\Delta_{H/L} = +0.5103$ (95% BCa CI: $[+0.226, +0.697]$, $p < 0.0001$).

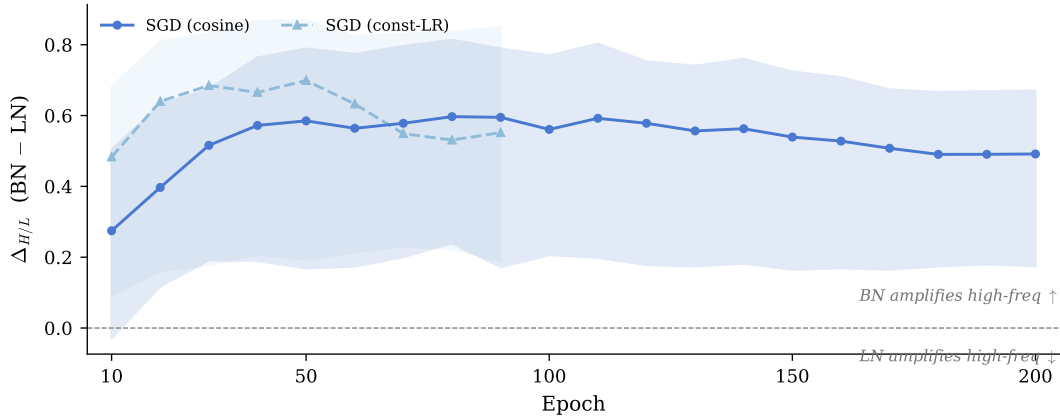


Figure 3: $\Delta_{H/L}$ trajectory under constant LR (green) overlaid on the cosine-schedule baseline (blue/orange). The constant-LR trajectory remains positive at all 20 checkpoints, peaking near $+0.70$ at epoch 50 and fluctuating within $[+0.51, +0.66]$ thereafter. This persistence confirms that the SGD-induced spectral gap is not an artifact of the cosine schedule.

Convergence asymmetry. Under constant LR, the BN–LN accuracy gap emerges early: BN reaches 31.0% by epoch 10 while LN reaches only 7.1%. The spectral gap $\Delta_{H/L}$ is positive from the earliest checkpoint ($+0.48$ at epoch 10, $p < 0.0001$). The cross-optimizer evidence from cosine-schedule experiments argues against interpreting positive $\Delta_{H/L}$ as merely a downstream consequence of the accuracy gap: AdamW produces a comparable BN–LN accuracy gap yet zero spectral divergence.

Cross-seed stability. The BN-side median post-normalization $r_{H/L}$ under constant LR is stable across seeds: s42 yields $\tilde{r}_{H/L}^{\text{BN}} = 0.564$ and s123 yields 0.715 (same order, same sign of $\Delta_{H/L}$). Both seeds produce a positive $\Delta_{H/L}$ at all 20 spectral checkpoints; the SGD–BN spectral bias under constant LR is therefore not seed-specific.

G.4 Weight Decay Sensitivity Analysis (W1a)

To isolate the effect of regularization strength on the normalization-induced spectral bias, we train ResNet-50 under SGD with two weight decay settings ($\lambda \in \{0.01, 0.001\}$) for both BN and LN, holding all other hyperparameters fixed ($\text{lr} = 0.01$, cosine schedule, 200 epochs, Tiny ImageNet). Table 14 reports the post-normalization median $r_{H/L}$ and the paired BN–LN gap $\Delta_{H/L}$ (BCa bootstrap, $B = 10,000$).

Reducing λ from 0.01 to 0.001 raises median $r_{H/L}$ for both normalization types, but the response is strikingly asymmetric. BN’s spectral profile is robust to the $10\times$ WD change (cross-WD

1211 $\Delta_{H/L} = -0.035$, $p = 0.064$, n.s.), consistent with low-pass filtering that is *intrinsic* to batch-level
1212 standardization rather than a side effect of regularization strength. LN shifts substantially (cross-
1213 WD $\Delta_{H/L} = -0.293$, $p < 0.001$); LN’s spectral profile is therefore *WD-contingent*. The BN–LN
1214 gap narrows at lower λ because LN’s median $r_{H/L}$ rises from 0.197 to 0.494, not because BN’s
1215 declines ($0.521 \rightarrow 0.687$); the gap reduction is driven entirely by the LN end. Although BN’s raw
1216 median $r_{H/L}$ increases from 0.521 to 0.687, the paired test captures per-layer consistency: the shift
1217 is driven by a small number of deep-stage layers, and the per-layer differences are not statistically
1218 significant ($p = 0.064$). At $\lambda=0.001$, the BN–LN gap remained directionally consistent (BN > LN)
1219 but did not reach significance ($\Delta_{H/L} = +0.078$, $p = 0.054$, 95% CI $[-0.000, +0.163]$).

1220 **Caveat.** The $\lambda=0.01$ LN configuration exhibits systematic underfitting (train accuracy $\approx 66\%$, val
1221 44.7%), so its larger paired gap ($\Delta_{H/L} = +0.277$, $p < 0.001$) is confounded by convergence quality.
1222 The $\lambda=0.001$ pair provides the fairer comparison. All W1a configurations use a single seed ($s=42$);
1223 multi-seed replication is deferred to future work.

Table 14: Weight decay sensitivity (W1a): post-normalization median $r_{H/L}$ and paired spectral gap $\Delta_{H/L}$ (BCa bootstrap, $B=10,000$) for ResNet-50+SGD on Tiny ImageNet. Top: paired BN–LN gap within each λ . Bottom: cross-WD change within each norm type. [†]Systematic underfitting (train acc $\approx 66\%$).

λ	Norm	Val Acc (%)	Median $r_{H/L}$	$\Delta_{H/L}$	95% CI	p
0.01	BN	58.39	0.521	+0.277	[+0.188, +0.558]	<0.001
0.01	LN	44.72 [†]	0.197			
0.001	BN	55.61	0.687	+0.078	[−0.000, +0.163]	0.054
0.001	LN	43.60	0.494			
<i>Cross-WD</i> (λ : 0.01 $\rightarrow$ 0.001)						
	BN			−0.035	[−0.109, −0.000]	0.064
	LN			−0.293	[−0.362, −0.222]	<0.001

1224 H LN+SGD Training Instability

1225 ResNet-50+SGD+LN exhibits seed-dependent training instability absent from the BN variant. Fig-
1226 ure 4 plots validation accuracy curves for all four seeds under both normalizations. BN converges
1227 consistently (57.97, 58.75, 58.86, 58.91%; $\sigma \approx 0.4$ pp), while LN shows high variance: seed 0 con-
1228 verges to 43.31%, seed 123 to 43.79%, seed 7 to 42.94%, and seed 42 collapses catastrophically
1229 at epoch 52, with validation accuracy falling from $\sim 16\%$ to chance level (0.50% for 200 classes)
1230 within a few epochs. The collapse is accompanied by train loss rising to $\ln(200) \approx 5.3$ (uniform
1231 class posterior), consistent with complete representational failure.

1232 This instability is specific to the LN+SGD combination on ResNet-50; BN+SGD with the same hy-
1233 perparameters, seeds, and hardware produces no instability. The failure is consistent with known
1234 sensitivity of LN-based CNNs to SGD’s uniform learning rate across heterogeneous parameter
1235 groups Ba et al. [2016], though the stochastic nature (only 1 of 4 seeds fails) suggests the insta-
1236 bility is a seed-specific basin-of-attraction problem rather than a systematic divergence.

1237 Because the seed 42 LN run does not produce a converged model, all spectral analyses for this
1238 configuration use the three surviving seeds (s0, s123, s7); the resulting BCa CIs yield a significant
1239 positive $\Delta_{H/L}$.

1240 **Survivorship bias disclosure.** Reporting results from three surviving seeds out of four attempted
1241 introduces potential survivorship bias: if the collapsed seed would have produced a qualitatively
1242 different spectral profile, the reported $\Delta_{H/L}$ may not represent the full distribution of outcomes. We
1243 mitigate this concern by noting: (1) the collapse occurred at epoch 52 before convergence, so no
1244 meaningful spectral estimate is extractable; (2) all three surviving seeds yield individually signifi-
1245 cant positive $\Delta_{H/L}$ with consistent sign and magnitude; (3) the collapsed seed’s partial trajectory
1246 (validation accuracy dropped from $\sim 38\%$ to chance level) is included in Figure 4 for transparency.

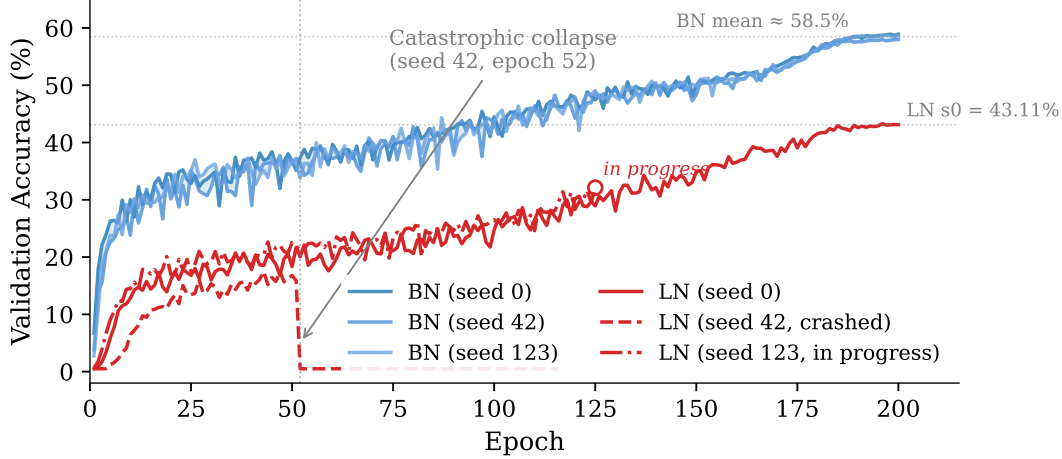


Figure 4: ResNet-50+SGD training curves under BN (blue) and LN (red), Tiny ImageNet. BN converges consistently across four seeds (shaded band: min–max). LN seed 0 converges to 43.31%, seed 123 to 43.79%, seed 7 to 42.94%, but seed 42 collapses at epoch 52. Arrow marks collapse onset.

I Block Collapse Extended Discussion

I.1 SGD Training Collapse Analysis

Positive $\Delta_{H/L}$ under SGD was confirmed only in ResNet-50 and EfficientNet-B0 (with warmup), both BN-native CNNs. SGD training of non-CNN architectures produces catastrophic block collapse in DeiT-S and ConvNeXt-T (both normalization-independent) and in EfficientNet-B0+LN without warmup (normalization-specific); with 5-epoch warmup, EfficientNet-B0+SGD converges and yields $\Delta_{H/L} = +0.253$ ($p = 0.003$). This result confirms SGD’s positive bias on a second CNN. VMamba-S+SGD yields a borderline negative $\Delta_{H/L} = -0.708$ (BH-adjusted $p = 0.049$) confounded by a ~ 14 pp accuracy gap; collapsed configurations are excluded from Table 1.

The block collapse observed under SGD for DeiT-S and ConvNeXt-T constitutes an independent finding with implications beyond spectral bias measurement. In DeiT-S, 17 of 25 layers (BN) and 18 of 25 layers (LN) produce near-zero spectral energy ($r_{H/L} < 0.01$). The collapse is therefore associated with the optimizer–architecture interaction rather than the normalization layer.

The contrasting survival patterns under BN and LN (blocks 0–2, an isolated layer at block 4, and the head norm for BN, totaling 8 layers, versus blocks 0–1 plus scattered deep layers for LN, totaling 7 layers) differ in topology but not in severity. Notably, Daneshmand et al. [2020] proved that BN provably avoids *rank* collapse at initialization; our results show this protection does not extend to training-time spectral collapse under SGD in Vision Transformers.

This observation connects to the well-documented difficulty of training Transformers with SGD Dong et al. [2021], Chen et al. [2022], Zhang et al. [2024b]. Dong et al. [2021] showed that self-attention layers suffer from rank collapse with depth; our finding differs in that the collapse we observe is optimizer-induced (SGD-specific) rather than architecture-inherent, and manifests in the frequency domain. The degradation intensifies sharply in late training: 17/25 layers are collapsed at the best checkpoint (epoch 199) versus 24/25 at epoch 200.

For the purposes of this study, block collapse renders $\Delta_{H/L}$ unmeasurable for the affected configurations, as the spectral ratio of collapsed layers is dominated by numerical noise. The positive spectral bias under SGD is therefore confirmed for ResNet-50 (and EfficientNet-B0 with warmup; Appendix E.1), where convolutional local inductive biases are compatible with SGD’s uniform update rule.

I.2 Random-Initialization Baseline

To test whether the BN–LN spectral gap requires training, we measure $\Delta_{H/L}$ on randomly initialized (untrained) models across four architectures, each with 10 random seeds (5 batches $\times$ 128 images per seed, `model.train()` so that BN computes batch statistics rather than using uninitialized running statistics). Table 15 reports the results.

Table 15: Spectral gap $\Delta_{H/L}$ at random initialization (10 seeds, t -test across seeds, `model.train()`). All four architectures show significant BN–LN spectral differences before any training ($p < 0.001$), but the optimizer–training interaction actively reshapes these initial biases.

Architecture	$\Delta_{H/L}^{\text{init}}$	std	p_{init}	H_0	$\Delta_{H/L}^{\text{trained}}$	Direction change
ResNet-50	+0.173	0.026	<0.001	Rejected	+0.454 (SGD)	amplified $\sim 2.6\times$
DeiT-S	+0.199	0.024	<0.001	Rejected	−0.227 (AdamW)	reversed
ConvNeXt-T	−1.548	1.041	0.001	Rejected	−0.320 (AdamW)	compressed $\sim 5\times$
EfficientNet-B0	+0.585	0.125	<0.001	Rejected	−0.369 (AdamW)	reversed

All four architectures exhibit significant BN–LN spectral differences at random initialization ($p < 0.001$); normalization layers therefore introduce a spectral bias before any training occurs. However, the post-training spectral gap is not a passive inheritance: SGD amplifies the initial positive bias in ResNet-50 ($\sim 2.6\times$), whereas AdamW reverses the sign in DeiT-S and EfficientNet-B0, and compresses ConvNeXt-T’s magnitude $\sim 5\times$. The spectral gap is thus actively shaped by the optimizer–training interaction, not merely set at initialization.

I.3 Learning-Rate–Aligned Control

To directly test the confound identified in the LINCHPIN analysis (that SGD and AdamW differ in both update rule *and* learning rate), we train ResNet-50+SGD at $\text{lr} = 0.001$ (matching AdamW’s default) with $\text{wd} = 0.01$ for 100 epochs on Tiny ImageNet ($s=42$). Both configurations converge fully (pre-registered threshold: train accuracy $> 15\%$; observed: BN 99.8%, LN 99.9%), removing the underfitting confound present in the W1a ablation’s $\text{wd} = 0.01$ pair.

Table 16: C1 learning-rate–aligned control: ResNet-50+SGD at $\text{lr} = 0.001$, $\text{wd} = 0.01$ (two seeds). The paired $\Delta_{H/L}$ is indistinguishable from zero in both seeds and when pooled; the spectral gap therefore requires high effective step size to manifest.

Configuration	Train acc	Val acc	Paired $\Delta_{H/L}$
<i>Seed 42 ($n = 25$)</i>			
SGD lr=0.001 BN	99.8%	57.4%	+0.024 $[-0.093, +0.056], p = 0.550$
SGD lr=0.001 LN	99.9%	50.0%	
<i>Seed 123 ($n = 25$)</i>			
SGD lr=0.001 BN	98.9%	57.3%	+0.022 $[-0.080, +0.087], p = 0.853$
SGD lr=0.001 LN	99.9%	49.4%	
Pooled ($n = 50$)	$\Delta_{H/L} = +0.023$, 95% CI $[-0.054, +0.056], p = 0.613$		

Table 17: Learning-rate gradient under SGD+BN/LN on ResNet-50 (Tiny ImageNet). The spectral gap scales monotonically with learning rate and vanishes at $\text{lr} = 0.001$.

lr	$\Delta_{H/L}$	95% CI	p	Source
0.1	+0.454	[+0.270, +0.498]	< 0.0001	Primary (Table 1)
0.01	+0.277	[+0.188, +0.558]	< 0.001	W1a (§G.4)
0.001	+0.023	[−0.054, +0.056]	0.613	C1 pooled $n=50$ (this section)

The monotonic decline from +0.454 to +0.023 over a two-order-of-magnitude lr reduction confirms that the spectral gap is driven by the effective update magnitude. At $\text{lr} = 0.001$ (the default for AdamW), SGD produces a $\Delta_{H/L}$ indistinguishable from AdamW’s (−0.016). The LINCHPIN effect is therefore primarily lr -driven rather than update-rule-driven. This conclusion is replicated across two seeds ($s \in \{42, 123\}$); both individual-seed and pooled ($n = 50$) paired tests yield $p > 0.5$.

I.4 AdamW Learning-Rate Gradient (MF3)

To complement the SGD lr-gradient (Table 17), we sweep learning rates under AdamW on ResNet-50 with BN and LN variants ($s=42$, Tiny ImageNet, 200 epochs).

Table 18: AdamW learning-rate gradient on ResNet-50 (MF3). Val accuracy is best-epoch validation accuracy. LN variants at $\text{lr} \geq 0.005$ collapse within epoch 1. Warmup: 5-epoch linear for $\text{lr} \geq 0.003$, standard cosine otherwise.

lr	BN val	LN val	LN best ep	$\Delta_{H/L}$	95% CI	p
0.0005	49.48%	37.68%	ep 189	+0.023	$[-0.263, +0.142]$	0.838
0.001	49.48%	37.68%	ep 189	-0.016	$[-0.128, +0.028]$	0.802
0.002	50.92%	37.06%	ep 200	+0.141	$[-0.128, +0.525]$	0.531
0.003	53.45%	39.18%	ep 190	+0.299	$[-0.014, +0.544]$	0.004
0.005	34.44% [†]	collapsed	—	—	—	—
0.01	23.33% [†]	collapsed	—	—	—	—

[†]Training incomplete (9 epochs only); BN had not converged.

At $\text{lr} = 0.005$ and 0.01 , the LN variant diverges within the first epoch (val accuracy $< 2\%$), while BN training was terminated after 9 epochs without convergence. The stability boundary lies between $\text{lr} = 0.003$ and 0.005 . At moderate learning rates, BN maintains a ~ 14 pp accuracy advantage over LN, comparable to the SGD baseline gap. LN converges slowly, reaching best validation accuracy near the end of training (epoch 190 at $\text{lr} = 0.003$, epoch 200 at $\text{lr} = 0.002$), with BN maintaining a consistent ~ 14 pp advantage throughout.

The inconsistent warmup schedule (5-epoch linear for $\text{lr} \geq 0.003$, standard cosine for lower lr) prevents quantitative cross-lr monotonicity claims; within each lr , the BN–LN comparison remains valid.

J Corruption Robustness

To probe whether the spectral bias documented in Table 1 has measurable downstream consequences, we evaluate ResNet-50 under four optimizer–normalization configurations on corrupted Tiny ImageNet inputs. We group ImageNet-C Hendrycks and Dietterich [2019] corruptions into high-frequency (Gaussian noise, shot noise, impulse noise) and low-frequency (fog, contrast, brightness) categories and report the average accuracy drop relative to clean performance for each group.

Table 19: Corruption robustness of ResNet-50 under four optimizer–normalization configurations on Tiny ImageNet. **HF drop**: mean accuracy drop on high-frequency corruptions. **LF drop**: mean accuracy drop on low-frequency corruptions. **HF–LF gap**: difference between the two drops (larger values indicate disproportionate high-frequency sensitivity). **Rel. rob.**: HF corrupted accuracy / LF corrupted accuracy (closer to 1 = more balanced sensitivity). $\Delta_{H/L}$ from Table 1 is reproduced for reference.

Configuration	Clean	HF drop	LF drop	HF–LF gap	Rel. rob.	$\Delta_{H/L}$
SGD+BN	58.9%	47.0	37.4	9.6	0.554	+0.454
SGD+LN	39.4%	32.0	20.7	11.3	0.400	—
AdamW+BN	49.4%	36.3	33.1	3.2	0.806	-0.016
AdamW+LN	37.7%	28.2	21.7	6.5	0.594	—

Spectral bias predicts corruption asymmetry. SGD+BN, the only configuration with a significant positive $\Delta_{H/L}$ (+0.454), exhibits a 9.6 pp gap between HF and LF accuracy drops, the largest among BN variants. The corresponding AdamW+BN configuration, whose $\Delta_{H/L} = -0.016$ is indistinguishable from zero, shows only a 3.2 pp gap: a $3\times$ reduction in the HF–LF asymmetry. This pattern is consistent with the spectral bias account: configurations that concentrate more energy at high frequencies (positive $\Delta_{H/L}$) are disproportionately sensitive to high-frequency corruptions. The relative robustness ratio corroborates this: AdamW configurations (0.806, 0.594) exhibit more balanced corruption sensitivity than their SGD counterparts (0.554, 0.400).

Caveats. These results are preliminary and should be interpreted with caution. First, the four configurations differ substantially in clean accuracy (39.4–58.9%), and absolute accuracy drops scale with baseline performance; the relative robustness ratio partially controls for this but is not a perfect correction. Second, the analysis is restricted to a single architecture (ResNet-50) on a single dataset (Tiny ImageNet), so the corruption–spectral link may not generalize to other architectures or scales. Third, the HF/LF corruption grouping is coarse: fog and contrast manipulations are not purely low-frequency, and the mapping from corruption type to spectral band is approximate. We report these results as suggestive evidence that the spectral bias measured by $\Delta_{H/L}$ has functional consequences for corruption robustness, not as a definitive causal claim.

J.1 Cross-Architecture Spectral Gap Under Corruption

To test whether the spectral bias documented in Table 1 persists under input corruptions, we evaluate $\Delta_{H/L}$ under five corruption types at three severity levels across four architecture–optimizer pairs (Table 20). ResNet-50+SGD was evaluated in an independent run (LN checkpoint unavailable in the main batch); results are consistent with the primary $\Delta_{H/L}$ estimate (+0.464 vs. +0.454, within the 95% CI of the primary analysis).

Table 20: Spectral gap $\Delta_{H/L}$ under input corruptions (BCa bootstrap, $B = 10,000$). *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$. ResNet-50+SGD from independent run[†]; all others from main C4 batch. ConvNeXt-T+AdamW ($n = 9$ layers) is underpowered: no condition reaches significance.

Condition	ResNet+SGD [†] ($n=25$)	ResNet+AdamW ($n=25$)	DeiT+AdamW ($n=25$)	ConvNeXt+AdamW ($n=9$)	EffNet+AdamW ($n=16$)
Clean	+0.464***	+0.008	−0.218***	−0.111	−0.351**
Gauss. noise s1	+0.466***	+0.010	−0.221***	−0.116	−0.373**
Gauss. noise s2	+0.467***	+0.009	−0.226***	−0.120	−0.398**
Gauss. noise s3	+0.474***	+0.010	−0.236***	−0.311	−0.445**
Shot noise s1	+0.474***	+0.010	−0.237***	−0.310	−0.443**
Shot noise s2	+0.486***	+0.005	−0.245***	−0.149	−0.525**
Shot noise s3	+0.481***	−0.001	−0.272***	−0.170	−0.616***
Gauss. blur s1	+0.373***	−0.093	−0.166***	−0.182	−0.299***
Gauss. blur s2	+0.315***	−0.077*	−0.137***	−1.968	−0.154***
Gauss. blur s3	+0.257***	−0.067	−0.111***	−7.551	−0.115***
Contrast s1	+0.366***	−0.062	−0.135***	−0.268	−0.332***
Contrast s2	+0.321***	−0.095	−0.104***	−0.241	−0.355***
Contrast s3	+0.201***	−0.148*	−0.055***	−0.163	−0.390***
Brightness s1	+0.463***	+0.005	−0.215***	−0.108	−0.351**
Brightness s2	+0.462***	+0.001	−0.213***	−0.104	−0.351**
Brightness s3	+0.460***	−0.005	−0.210***	−0.101	−0.351**
<i>Summary</i>					
Sign consistency	16/16 pos.	mixed	16/16 neg.	16/16 neg.	16/16 neg.
$\Delta_{H/L}$ range	[+0.201, +0.486]	[−0.148, +0.010]	[−0.272, −0.055]	[−7.551, −0.101]	[−0.616, −0.115]
All $p <$	0.001	n.s.	0.001	n.s.	0.002

Key patterns. For all configurations with significant baseline spectral gaps, sign is preserved under all 16 conditions (clean + 5 corruptions $\times$ 3 severities): ResNet-50+SGD (16/16 positive, all $p < 0.001$), DeiT-S+AdamW (16/16 negative, all $p < 0.001$), and EfficientNet-B0+AdamW (16/16 negative, all $p < 0.002$). Additive noise (Gaussian, shot) amplifies $|\Delta_{H/L}|$, while frequency-attenuating corruptions (blur, contrast) reduce it. This pattern is consistent with noise amplifying the frequency-band processing difference between BN and LN. ResNet-50+AdamW, whose baseline is near zero (+0.008, n.s.), remains near zero under all corruptions; the null spectral gap is equally stable. ConvNeXt-T+AdamW retains only 9 layers and does not reach significance under any condition.

K Sensitivity and Robustness Analyses

K.1 Cutoff Sensitivity Analysis

Our main analysis filters layers with spatial resolution below 8×8 (§3.1). To verify that the reported findings are robust to this choice, we recompute $\Delta_{H/L} = \text{median}(\Delta_{H/L})$ under spatial cutoffs ranging from 4×4 to 16×16 . Table 21 reports the results.

Table 21: Cutoff sensitivity analysis. Each cell shows $\Delta_{H/L}$ and the number of eligible layers n . Significance ($p < 0.05$, 95% BCa CI excludes zero) is marked with *. A dash (–) indicates no layers survive the cutoff. Cutoffs ≤ 8 yield identical results because the spectral extraction pipeline enforces a minimum spatial size of 8×8 . Point estimates may differ slightly from Table 1 due to different seed pooling (not BCa bias correction, which affects only CIs). [†]Uses independent bootstrap ($n_{\text{BN}}=50$, $n_{\text{LN}}=25$) after excluding the crashed LN seed 42.

Configuration	$c = 4$	$c = 8$ (default)	$c = 10$	$c = 16$
ResNet-50+SGD (TinyIN) [†]	+0.446* (50)	+0.446* (50)	+0.193* (24)	+0.193* (24)
ResNet-50+SGD (IN100)	+0.398* (44)	+0.398* (44)	+0.398* (44)	+0.223* (25)
ResNet-50+AdamW (TinyIN)	−0.016 (50)	−0.016 (50)	−0.145 (24)	−0.145 (24)
ResNet-50+AdamW (IN100)	+0.010 (44)	+0.010 (44)	+0.010 (44)	+0.155 (25)
DeiT-S+AdamW (TinyIN)	−0.227* (25)	−0.227* (25)	–	–
ConvNeXt-T+AdamW (TinyIN)	−0.320* (27)	−0.320* (27)	−0.320* (15)	−0.320* (15)
VMamba-S+AdamW (TinyIN)	−7.742* (28)	−7.742* (28)	−10.598* (14)	−10.598* (14)
MambaVision-T+AdamW (TinyIN)	−35.194* (12)	−35.194* (12)	−0.943 (4)	−0.943 (4)

All eight configurations maintain consistent $\Delta_{H/L}$ signs across every tested cutoff. Stricter thresholds reduce the number of eligible layers: DeiT-S has no layers above 8×8 (due to its 8×8 patch embeddings on 64×64 input), and MambaVision-T retains only 4 layers at $c \geq 10$. These architectures lose statistical power but preserve the negative sign. The SGD-induced positive bias and the AdamW-induced non-positive pattern are fully robust to the choice of spatial resolution threshold.

MambaVision-T magnitude sensitivity. While the *sign* of MambaVision-T’s $\Delta_{H/L}$ is stable across cutoffs, its *magnitude* is highly sensitive: $\Delta_{H/L}$ drops from -35.194 ($n = 12$, $c \leq 8$) to -0.943 ($n = 4$, $c \geq 10$), a 97% reduction. This indicates that the extreme spectral gap is concentrated in the lowest-resolution layers (8×8), which dominate the median at the default cutoff. The magnitude claims for MambaVision-T in the main text (§5, Appendix M.1) should therefore be interpreted as cutoff-dependent; the directional finding (negative $\Delta_{H/L}$) is robust.

K.2 Sensitivity to Collapse Threshold ε

We vary the collapse threshold $\varepsilon \in \{0.01, 0.05, 0.10, 0.15, 0.20\}$ (layers with $r_{H/L} < \varepsilon$ are excluded from $\Delta_{H/L}$ computation). All five primary AdamW configurations retain their non-positive $\Delta_{H/L}$ sign at every tested ε ; the cross-configuration sign test ($m = 5$, $p = 0.031$) is unaffected. The only sign flip occurs for DeiT-S+SGD at $\varepsilon \geq 0.05$ ($-0.066 \rightarrow +0.073$), but this configuration is non-significant at all thresholds ($p > 0.68$, $n \leq 15$) and is excluded from Table 1 due to block collapse. Four architectures (ConvNeXt-T, VMamba-S, MambaVision-T, EfficientNet-B0+AdamW) show *identical* $\Delta_{H/L}$ across all ε ; no layers in these architectures fall in the $[0.01, 0.20]$ range. The default $\varepsilon = 0.01$ is thus a conservative choice that does not bias the directional conclusions.

K.3 Effective Sample Size Adjustment

To address potential narrowing of bootstrap CIs due to inter-layer correlation, we employ two correction methods. Our **primary correction** is a stage-level block bootstrap that respects the grouped structure of layers within architectural stages and produces robust CIs without parametric assumptions about correlation decay. Blocks correspond to architectural depth stages (e.g., 4 stages for ResNet-50, each containing 3–16 normalization layers depending on stage depth), so block size varies by architecture rather than being fixed; at each bootstrap iteration, entire stages are resampled with replacement (n_{stages} draws from n_{stages} available). As a conservative sensitivity check, we also compute n_{eff} via the initial positive sequence estimator Geyer [1992] applied to the lag-1 autocorrelation ρ_1 of layer-level BN–LN difference vectors ordered by depth. The n_{eff} -based design effect

ranged from 1.00 (architectures with negative ρ_1 , where naive CIs are already conservative) to 6.88 (ResNet+AdamW on IN100, $\rho_1=0.65$), with CI expansion factors from $1.00\times$ to $2.63\times$. The two methods give concordant results (mean CI expansion $1.32\times$ block vs $1.55\times$ n_{eff}).

Under block bootstrap CIs, all primary conclusions are preserved. Two configurations become non-significant only under the more conservative n_{eff} adjustment: (i) MambaVision+AdamW, where $n_{\text{eff}} = 6.1$ from $n = 12$ layers may overpenalize given the small baseline sample (the block bootstrap CI $[-36.0, -2.07]$ remains clearly significant); (ii) VMamba+SGD, which is the most fragile configuration under any correction method and should be interpreted with caution. ResNet+SGD on TinyImageNet remains significant even after n_{eff} adjustment ($[+0.14, +0.53]$; 3-seed pooled, $n=75$, $\rho_1=0.37$, $n_{\text{eff}}=26.0$). Architectures showing negative ρ_1 (DeiT+AdamW, ConvNeXt+AdamW, VMamba+AdamW) require no correction.

LN seed attrition. For ResNet-50+SGD, LN seed 42 collapsed at epoch 52 (Appendix H), but the three surviving LN seeds (s0, s123, s7) achieve the target 3-seed pooling ($n = 25 \times 3 = 75$ layer pairs), matching the BN side. The resulting CI $[+0.270, +0.498]$ does not cross zero ($p < 0.0001$), and the block bootstrap (above) confirms significance under dependence-corrected resampling.

K.4 BN Leave-One-Out Seed Sensitivity

Table 22 reports BN leave-one-out CI widths for ResNet-50+SGD (TinyImageNet). Dropping one BN seed widens the CI by $1.03\text{--}1.17\times$; the LN three-seed width (0.076) is narrower than any BN three-seed combination. Recomputing pooled paired $\Delta_{H/L}$ after each BN seed removal yields a range of $[+0.438, +0.471]$ ($< 10\%$ of $+0.454$); the estimate is therefore stable.

Table 22: BN leave-one-out seed sensitivity (ResNet-50+SGD, TinyImageNet). CI width expansion from dropping one BN seed ranges from $1.03\times$ to $1.17\times$. LN three-seed CI width is shown for reference.

Configuration	n	Median $r_{H/L}$	95% CI	Width
BN 4-seed (s0+s42+s123+s7)	100	+0.635	$[+0.533, +0.769]$	0.236
drop s0 (s42+s123+s7)	75	+0.650	$[+0.512, +0.769]$	0.257 ($1.09\times$)
drop s42 (s0+s123+s7)	75	+0.625	$[+0.513, +0.784]$	0.271 ($1.15\times$)
drop s123 (s0+s42+s7)	75	+0.625	$[+0.512, +0.755]$	0.243 ($1.03\times$)
drop s7 (s0+s42+s123)	75	+0.620	$[+0.493, +0.769]$	0.276 ($1.17\times$)
LN 3-seed (s0+s123+s7)	75	+0.179	$[+0.140, +0.216]$	0.076

K.5 Frequency Cutoff Sensitivity (ρ_{cut})

The frequency cutoff ρ_{cut} defines the boundary between “high” and “low” frequency in the 2D FFT power spectrum: $r_{H/L} = E(\rho > \rho_{\text{cut}})/E(\rho \leq \rho_{\text{cut}})$. Our main analysis uses $\rho_{\text{cut}} = f_N/2$ (half the Nyquist frequency). To verify that the reported $\Delta_{H/L}$ findings are robust to this choice, we recompute paired BCa bootstrap $\Delta_{H/L}$ at four cutoffs spanning the range $[f_N/4, 2f_N/3]$. All cutoffs are pre-computed from the exact 2D FFT power spectrum. Table 23 reports the results.

The sign of $\Delta_{H/L}$ is preserved across all four cutoffs for 9 of 10 configurations. All seven configurations that are significant at the default cutoff ($f_N/2$) remain significant across the full range for purely convolutional and transformer architectures; the SGD-positive/AdamW-negative pattern is fully robust to frequency cutoff choice.

Magnitude scaling. $|\Delta_{H/L}|$ monotonically decreases as ρ_{cut} increases: a lower cutoff places more of the spectrum in the “high” frequency bin, amplifying the ratio and thus the BN–LN difference. This is expected and does not affect sign or significance.

ResNet+AdamW significance gain at low cutoffs. ResNet+AdamW (TinyIN) is non-significant at the default cutoff ($p = 0.807$) but becomes significant at $\rho = f_N/4$ ($p = 0.005$). The sign is consistently negative across all cutoffs ($\Delta_{H/L} \in [-0.169, -0.013]$), consistent with the broader AdamW pattern.

Table 23: Frequency cutoff sensitivity. Each cell shows $\Delta_{H/L}$ (BCa $B=10,000$). *: significant at $\alpha = 0.05$ (95% CI excludes zero). The default cutoff $f_N/2$ corresponds to the main analysis (Table 1).

Configuration	$\rho = f_N/4$	$\rho = f_N/3$	$\rho = f_N/2$ (default)	$\rho = 2f_N/3$
ResNet+SGD (TinyIN)	+1.081*	+1.045*	+0.454*	+0.254*
ResNet+SGD (IN100)	+1.066*	+0.615*	+0.357*	+0.151*
DeiT+AdamW (TinyIN)	-0.515*	-0.515*	-0.227*	-0.143*
DeiT+AdamW (IN100)	-1.214*	-0.565*	-0.424*	-0.177*
EfficientNet+SGD	+0.803*	+0.767*	+0.260*	+0.109*
EfficientNet+AdamW	-1.700*	-1.380*	-0.369*	-0.187*
ConvNeXt+AdamW	-1.323*	-0.843*	-0.321*	-0.168*
ResNet+AdamW (TinyIN)	-0.169*	-0.167*	-0.016	-0.013
VMamba+AdamW	-82.190	-30.091	-7.742*	-4.640*
MambaVision+AdamW	+105.342	+142.094	-35.194*	-11.687

1420 **SSM architectures at low cutoffs.** VMamba-S and MambaVision-T exhibit extreme variance at
1421 $\rho \leq f_N/3$, with MambaVision-T showing a sign flip (+105 vs. -35 at the default). This instability
1422 arises because SSM feature maps concentrate energy very close to DC; a cutoff near DC creates
1423 a tiny “low-frequency” bin dominated by numerical noise. At the default and higher cutoffs ($\rho \geq$
1424 $f_N/2$), both architectures show robustly negative $\Delta_{H/L}$.

1425 K.6 Pre-Normalization Spectral Analysis

1426 To probe whether normalization layers act as the primary spectral filter or whether the optimizer
1427 already shapes frequency content before normalization, we extract $r_{H/L}$ from the activations *imme-*
1428 *diately before* each normalization layer.

Table 24: Pre- vs. post-normalization spectral gap for key configurations.

Configuration	$\Delta_{H/L}^{\text{pre}}$	p_{pre}	$\Delta_{H/L}^{\text{post}}$	p_{post}
ResNet+SGD	+0.151	0.0002	+0.454	<0.001
ResNet+AdamW	+0.180	<0.001	-0.016	0.81
DeiT+AdamW	-0.708	<0.001	-0.227	<0.001
ConvNeXt+AdamW	-0.425	0.067	-0.320	0.029

1429 Pre-normalization representations already exhibit optimizer-dependent spectral bias ($\Delta_{H/L}^{\text{pre}} \neq 0$,
1430 $p < 0.001$ for three of the four configurations; ConvNeXt+AdamW is marginally non-significant
1431 with $p = 0.067$), rejecting the hypothesis that normalization alone creates the spectral gap. How-
1432 ever, normalization does not simply pass through the pre-existing pattern: (1) Under SGD, batch
1433 normalization *amplifies* the BN>LN spectral difference by $3.0\times$ (+0.151 $\rightarrow$ +0.454); (2) Under
1434 AdamW, batch normalization *erases* a pre-existing positive difference (+0.180 $\rightarrow$ -0.016, non-
1435 significant); (3) For DeiT+AdamW, normalization *attenuates* the pre-existing negative pattern by
1436 $3.1\times$ (-0.708 $\rightarrow$ -0.227); (4) For ConvNeXt+AdamW, normalization *preserves* the negative di-
1437 rection with modest attenuation (-0.425 $\rightarrow$ -0.320, $p_{\text{post}} = 0.029$).

1438 These results demonstrate that both optimizer and normalization contribute to the observed spectral
1439 signature. The optimizer sets the baseline spectral bias of the learned representation; normalization
1440 acts as a frequency-selective modulator whose effect depends on the (architecture, optimizer, norm-
1441 type) triple.

1442 **ConvNeXt-T pre-normalization CI.** ConvNeXt-T+AdamW exhibits a wide pre-normalization
1443 bootstrap CI ($\Delta_{H/L}^{\text{pre}} = -0.425$, 95% CI [-12.75, +0.10], $p = 0.067$, $n = 27$), analogous to the
1444 wide post-normalization CI discussed in Appendix C.4. The CI breadth is likely driven by the same
1445 stage-3 outlier layers identified in the post-normalization analysis; after trimming in that setting, the
1446 point estimate remained stable at $\Delta_{H/L} = -0.320$ with a 39% narrower CI. Although per-layer
1447 pre-normalization data are not available for formal trimming, the negative point estimate direction
1448 is consistent across both pre- and post-normalization views, and the marginally non-significant p -

value (0.067) reflects limited statistical power from the small effective sample ($n = 27$) rather than absence of effect.

K.7 Spatial Resolution Filter: Per-Architecture Details

The spatial resolution filter (§3.1) excludes all normalization layers whose feature maps fall below 8×8 . By the Nyquist–Shannon sampling theorem Shannon [1949], an $N \times N$ feature map yields at most $\lfloor N/2 \rfloor$ independent frequency bins per axis after the 2D DFT. When $N < 8$, fewer than 4 radial bins remain, and the low/high partition at $\rho_{\text{cut}} = \lfloor N/4 \rfloor$ spans at most 1–2 bins per band, too coarse for a meaningful $r_{H/L}$ estimate. The fraction of retained layers varies from 21% to 100% across architectures, depending on each model’s downsampling schedule. Table 25 reports the per-architecture breakdown on Tiny ImageNet (64×64).

Table 25: Spatial resolution filter outcomes per architecture on Tiny ImageNet (64×64). **Total:** normalization layers in the model. **Valid** ($\geq 8 \times 8$): layers surviving the filter.

Architecture	Total	Valid ($\geq 8 \times 8$)	Retention
DeiT-S	25	25	100%
ResNet-50	53	25	47%
ConvNeXt-T	23	9	39%
EfficientNet-B0	49	16	33%
VMamba-S	50	14	28%
MambaVision-T	29	6	21%

Per-architecture downsampling notes. *DeiT-S* retains all 25 normalization layers because the 64×64 input is divided into non-overlapping 8×8 patches at the stem, producing 8×8 feature maps that persist through all transformer blocks without further spatial reduction. At higher resolution (224×224 on ImageNet-100), each patch maps to the same 8×8 token grid (with 14 tokens per axis at patch size 16), so retention remains at 100%.

ResNet-50 uses four residual stages with $2 \times$ spatial downsampling at each stage transition. On 64×64 input, feature map resolutions progress through 32×32 (stage 1), 16×16 (stage 2), 8×8 (stage 3), and 4×4 (stage 4); normalization layers in stage 4 are excluded. On ImageNet-100 (224×224), stage 4 produces 7×7 feature maps that are also excluded, but stages 1–3 collectively contribute 44 valid layers (vs. 25 on Tiny ImageNet) due to the larger spatial extent of early stages.

ConvNeXt-T follows a 4-stage hierarchical design with one normalization layer per block (23 total). Its downsampling schedule mirrors ResNet but with $4 \times$ spatial reduction at the stem ($64 \rightarrow 16$), placing stages 3–4 below the 8×8 threshold and leaving 9 valid layers in stages 1–2.

EfficientNet-B0 contains 49 normalization layers across its depthwise-separable convolution (MB-Conv) blocks. Its compound scaling Tan and Le [2019] progressively reduces spatial resolution through 7 stages; 16 layers in the early stages retain $\geq 8 \times 8$ resolution on 64×64 input.

VMamba-S distributes 50 normalization layers across 4 hierarchical stages. The state-space model’s downsampling schedule mirrors that of hierarchical vision transformers; deeper stages operate at low spatial resolution, leaving 14 layers (28%) above the filter threshold.

MambaVision-T, a CNN–SSM hybrid, retains the fewest layers (6 of 29, 21%). Aggressive downsampling in the CNN stem and early stages reduces feature maps below 8×8 by the third stage. The extreme spectral gap magnitude ($\Delta_{H/L} = -35.194$) reported for this architecture is concentrated in these lowest-resolution surviving layers; cutoff sensitivity analysis (Appendix K.1) shows that the magnitude drops 97% when the threshold is raised to 10×10 , though the negative sign is preserved.

Resolution dependence of sample size. Because the filter operates on spatial resolution rather than network depth, the effective sample size n per configuration depends jointly on the architecture and the input resolution. This interaction is visible in the ResNet-50 comparison: $n = 25$ on Tiny ImageNet (64×64) versus $n = 44$ on ImageNet-100 (224×224), a $1.8 \times$ increase from the same architecture at higher input resolution. BCa CI width scales as $1/\sqrt{n}$ (Appendix C), so architectures with few surviving layers (MambaVision-T, $n = 6$) produce CIs roughly $2 \times$ wider than those with

many (DeiT-S, $n = 25$), a factor that should be considered when interpreting non-significant results for low-retention architectures.

K.8 Spectral Decomposition: Full Derivation

This section provides the complete mathematical derivation of the radial power profile and frequency band partition summarized in §3.1.

2D DFT and power spectrum. Given a single-channel feature map $\mathbf{F}_c \in \mathbb{R}^{H \times W}$ (the c -th channel of the C -channel tensor $\mathbf{F}$), its 2D discrete Fourier transform is:

$$\hat{\mathbf{F}}_c(u, v) = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} \mathbf{F}_c(x, y) e^{-2\pi i(ux/H + vy/W)}, \quad (5)$$

for frequency indices $u \in \{0, \dots, H-1\}$, $v \in \{0, \dots, W-1\}$. The power spectrum is $P_c(u, v) = |\hat{\mathbf{F}}_c(u, v)|^2$.

Radial binning. We shift the DC component to the center and compute the radial distance of each frequency component from the origin:

$$\rho(u, v) = \sqrt{(u - \lfloor H/2 \rfloor)^2 + (v - \lfloor W/2 \rfloor)^2}. \quad (6)$$

Radial bins are defined as integer-width annuli: bin k collects all (u, v) pairs with $\lfloor \rho(u, v) \rfloor = k$. The channel-averaged radial power at bin k is:

$$\bar{P}(k) = \frac{1}{C} \sum_{c=1}^C \frac{1}{|B_k|} \sum_{(u,v) \in B_k} P_c(u, v), \quad (7)$$

where $B_k = \{(u, v) : \lfloor \rho(u, v) \rfloor = k\}$ and $|B_k|$ is its cardinality. For an $N \times N$ feature map, the maximum representable radial frequency is $\rho_{\max} = \lfloor N/2 \rfloor$ (Nyquist limit); this produces $\lfloor N/2 \rfloor + 1$ radial bins (including the DC bin at $k = 0$).

Frequency band partition. The cutoff frequency $\rho_{\text{cut}} = \lfloor N/4 \rfloor$ bisects the radial frequency axis at half the Nyquist frequency:

$$E_L = \sum_{k=0}^{\rho_{\text{cut}}} \bar{P}(k), \quad E_H = \sum_{k=\rho_{\text{cut}}+1}^{\lfloor N/2 \rfloor} \bar{P}(k), \quad (8)$$

and the high-to-low frequency ratio is $r_{H/L} = E_H/E_L$ (Eq. 2).

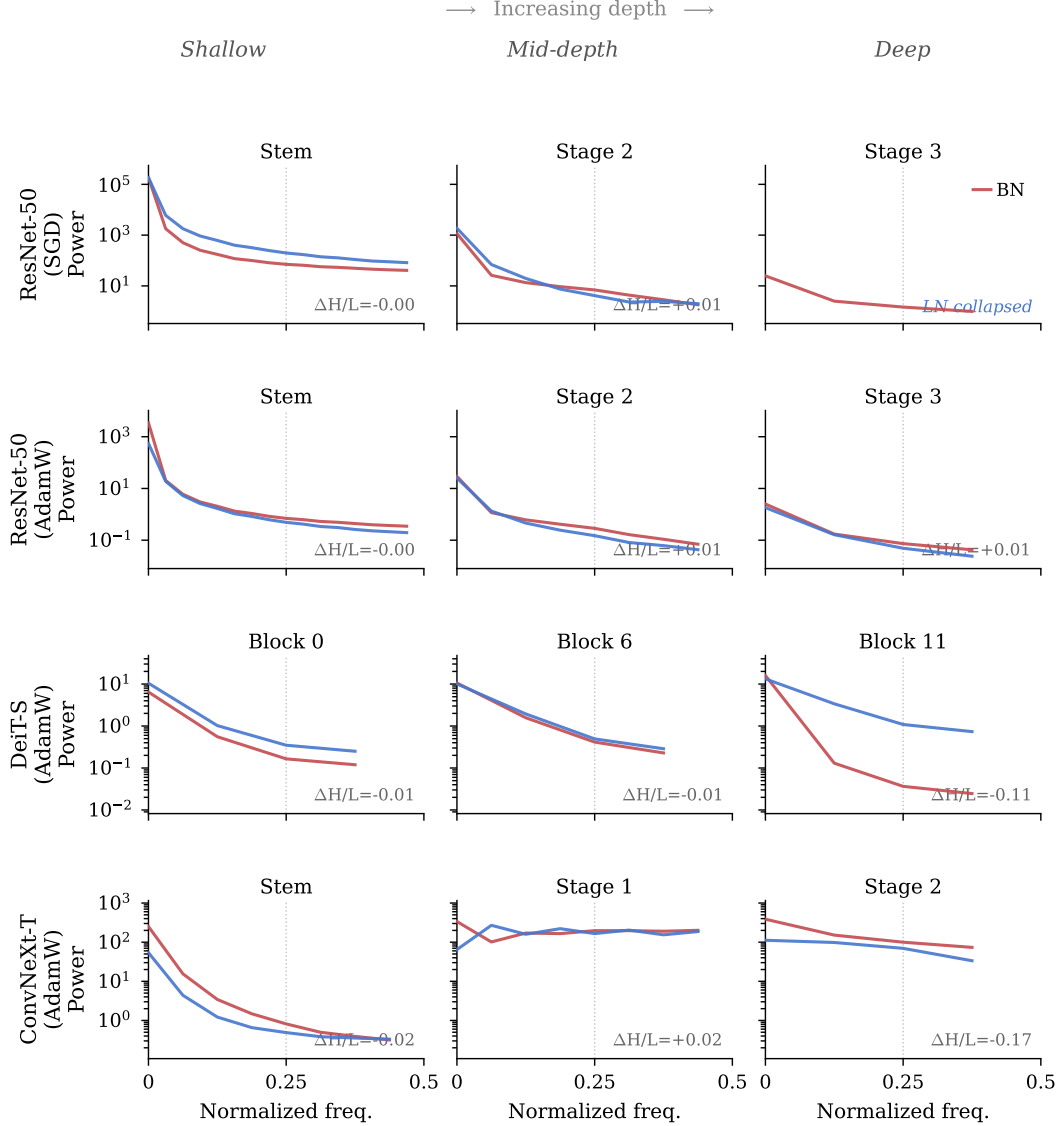
Minimum resolution constraint. When $N < 8$, the number of radial bins per band falls below 4 (specifically, $\rho_{\text{cut}} = \lfloor N/4 \rfloor \leq 1$ for $N \leq 7$), so the low-frequency band contains at most the DC component and one additional bin. In this regime, $r_{H/L}$ is dominated by quantization noise rather than genuine spectral structure. The $N \geq 8$ threshold ensures at least $\rho_{\text{cut}} = 2$ bins in the low band and $\lfloor N/2 \rfloor - \rho_{\text{cut}} + 1 \geq 2$ bins in the high band. This guarantees sufficient resolution for a stable partition. Table 25 reports the per-architecture impact of this constraint.

L Additional Figures

M Extended Discussion

M.1 Extended Implications

Our results carry several implications, each with a testable hypothesis.



$\Delta H/L$ values are per-layer differences, not pooled bootstrap estimates ($\hat{\theta}$).

Figure 5: Per-layer radial power spectra under BN (red) and LN (blue) at three network depths. Row 1: ResNet-50 under SGD, with clear BN–LN divergence and LN collapse at depth. Rows 2–4: ResNet-50, DeiT-S, and ConvNeXt-T under AdamW, where BN and LN spectra nearly overlap. Per-panel $\Delta_{H/L}$ values are single-layer differences, not the pooled bootstrap $\Delta_{H/L}$ in Table 1. VMamba-S and MambaVision-T are omitted due to $\sim 100\times$ magnitude difference in baseline $r_{H/L}$ (§5).

1518 **A frequency-domain account of texture bias.** Geirhos et al. [2019] demonstrated
1519 that ImageNet-trained CNNs exhibit a strong texture bias. Our finding that SGD+BN is the *only* con-
1520 figuration producing positive $\Delta_{H/L}$ provides a frequency-domain perspective: batch normalization
1521 under SGD amplifies high-frequency components, creating a representational prior toward texture.
1522 Under AdamW, this amplification disappears. **Hypothesis:** Replacing BN with LN in SGD-trained
1523 ResNets may reduce texture bias; switching from SGD to AdamW while retaining BN may produce
1524 a similar effect.

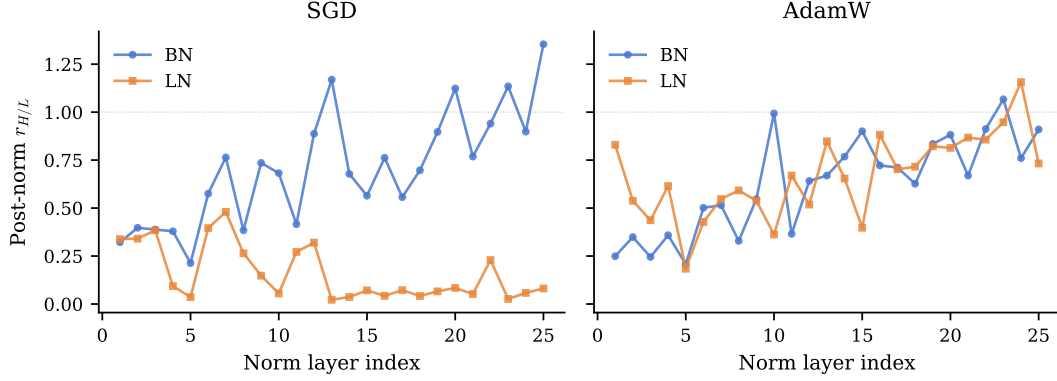


Figure 6: Per-layer post-normalization $r_{H/L}$ under BN and LN for ResNet-50 trained with SGD (left) and AdamW (right) on Tiny ImageNet (seed 42). Under SGD, BN consistently yields higher $r_{H/L}$ across layers; under AdamW, the two normalization schemes produce nearly indistinguishable spectral profiles.

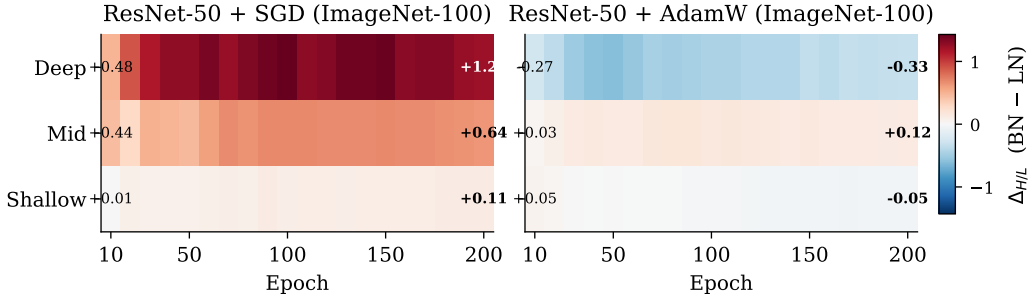


Figure 7: Depth-wise $\Delta_{H/L}$ decomposition for ResNet-50 on ImageNet-100 (seeds 42 and 123 pooled). SGD (left) produces positive $\Delta_{H/L}$ that increases with depth; AdamW (right) shows sign reversal in deep layers. Each row corresponds to a normalization layer ordered by network depth; columns show training epochs.

1525 **Spectral diagnostics for optimizer–normalization co-design.** The strong coupling implies that
 1526 $\Delta_{H/L}$ can serve as a diagnostic for optimizer–normalization compatibility. **Hypothesis:** For tasks
 1527 requiring high-frequency sensitivity (e.g., medical imaging), SGD+BN may yield higher task-
 1528 relevant feature preservation than AdamW+BN. For tasks requiring robustness to high-frequency
 1529 perturbations, AdamW+LN may be preferable.

1530 **Architecture-dependent magnitude demands multi-factor models.** The two-order-of-
 1531 magnitude variation in $|\Delta_{H/L}|$ demonstrates that the optimizer–normalization interaction does not
 1532 operate in isolation: the architecture’s spatial mixing mechanism shapes the magnitude. This rules
 1533 out single-factor accounts. **Hypothesis:** We conjecture that emerging architectures would benefit
 1534 from architecture-specific normalization strategies.

1535 **Implications for neural architecture search.** NAS evaluations implicitly carry spectral priors
 1536 inherited from the optimizer–normalization pairing. **Hypothesis:** A NAS search space that jointly
 1537 explores normalization type and optimizer may discover configurations with systematically different
 1538 spectral profiles than one that fixes the optimizer.

1539 **Spectral bias propagation in transfer learning.** The standard ImageNet pretraining pipeline uses
 1540 SGD+BN on ResNet, the only configuration producing positive $\Delta_{H/L}$. Pretrained representations
 1541 thus carry an embedded high-frequency bias before downstream adaptation. **Hypothesis:** Fine-

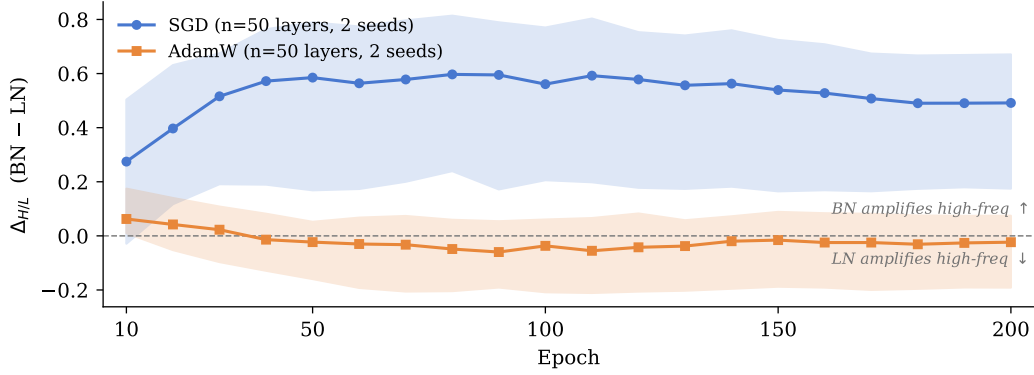


Figure 8: ResNet-50 $\Delta_{H/L}$ trajectory (seeds 42 and 123, Tiny ImageNet). SGD (blue) produces a $+0.27$ spectral gap by epoch 10, reaches $+0.57$ by epoch 40, plateaus near $+0.60$ through epoch 80, and converges to $+0.49$ (epoch-200 layer-wise median; see Table 1 for BCa bootstrap estimate) by epoch 200. AdamW (orange) remains within ± 0.06 throughout. Shading: IQR across 50 paired deltas ($25 \text{ layers} \times 2 \text{ seeds}$).

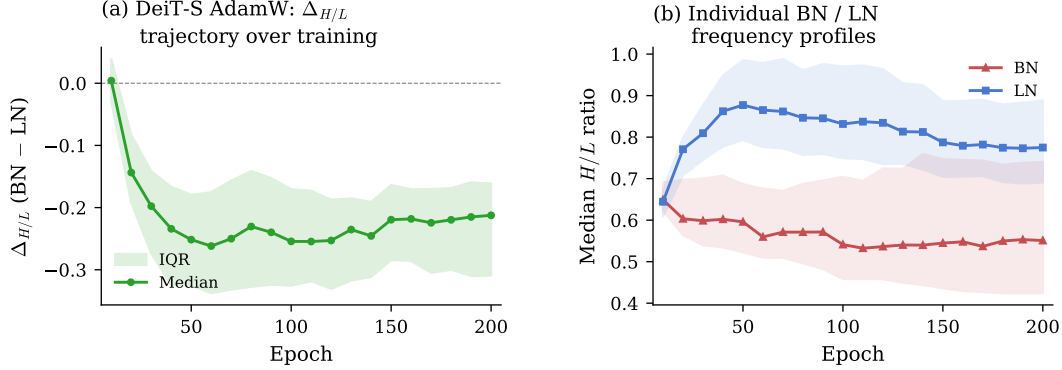


Figure 9: DeiT-S+AdamW $\Delta_{H/L}$ trajectory (seeds 42 and 0, Tiny ImageNet). (a) $\Delta_{H/L}$ remains negative throughout training, converging to -0.212 (epoch-200 layer-wise median; see Table 1 for BCa bootstrap estimate) by epoch 200. (b) Individual BN and LN median $r_{H/L}$ values diverge early: LN maintains higher spectral energy than BN across all epochs. Shading: IQR across 25 layer pairs per seed.

1542 tuning an SGD+BN-pretrained ResNet with AdamW may produce a measurable shift in $\Delta_{H/L}$
1543 toward zero.

1544 M.2 CIFAR-100 Overfitting and Implicit Regularization

1545 CIFAR-100 experiments expose a factorial interaction between normalization type, optimizer, and
1546 generalization. All four ResNet-50 configurations reach near-perfect training accuracy ($\geq 99.5\%$),
1547 yet validation performance diverges sharply: SGD+BN 77.6%, SGD+LN 76.1%, AdamW+BN
1548 75.7%, and AdamW+LN 67.9%. The train-val gap grows monotonically along two axes (replacing
1549 BN with LN, and replacing SGD with AdamW), reaching 32.1 pp for AdamW+LN versus 22.4 pp
1550 for SGD+BN. The normalization axis is consistent with BN’s implicit regularization via mini-batch
1551 noise Ioffe and Szegedy [2015], Luo et al. [2019]: BN injects stochasticity through batch-dependent
1552 statistics that LN, computing moments per-sample, cannot provide. The optimizer axis compounds
1553 this: AdamW’s per-parameter adaptive scaling may facilitate sharper minima and memorization
1554 on a 50K-sample dataset, an effect that BN’s noise partially counteracts but LN leaves unchecked.
1555 A parallel observation in EfficientNet-B0+AdamW+LN on Tiny ImageNet (~ 60 pp train-val gap)

suggests this vulnerability is not architecture-specific, though with only two instances we refrain from claiming generality. For practitioners, these results imply that substituting LN for BN without compensatory regularization (e.g., stronger dropout, weight decay, or data augmentation) risks severe overfitting, particularly under adaptive optimizers and limited data regimes. Because spectral extraction uses the best-validation checkpoint (§3.3), overfitting does not compromise the reported $\Delta_{H/L}$ values; however, it underscores that normalization’s implicit effects on training dynamics, beyond spectral bias, carry practical consequences for model selection.

M.3 Cross-Resolution Spectral Modulation

The cross-dataset comparison reveals a resolution-dependent modulation of spectral bias rather than simple sign replication. At 32×32 resolution, both optimizer conditions deviate from their higher-resolution patterns: SGD’s positive $\Delta_{H/L}$ collapses from +0.454 (Tiny ImageNet) and +0.357 (ImageNet-100) to a non-significant +0.032 ($p = 0.879$), while AdamW shifts from non-significant values (−0.016 on Tiny ImageNet, +0.010 on ImageNet-100) to a significantly positive +0.093 ($p = <0.0001$). Both deviations are consistent with a qualitatively different frequency regime at 32×32 : the maximum representable spatial frequency is limited to 16 cycles, compressing the frequency axis into a domain where BN–LN differences in high-frequency preservation are statistically unreliable. Because this low-resolution regime does not preserve the directional patterns observed at higher resolutions, we report CIFAR-100 $\Delta_{H/L}$ values separately from the primary BH-corrected table (Table 1).

Spectral resolution and statistical precision at 32×32 . The quantitative impact of resolution on spectral measurement precision is governed by bin cardinality in the discrete frequency domain Shannon [1949]. At 32×32 , the Nyquist limit is $\rho_{\max} = \lfloor 32/2 \rfloor = 16$ cycles and the half-Nyquist cutoff is $\rho_{\text{cut}} = \lfloor 16/2 \rfloor = 8$, partitioning the spectrum into 9 low-frequency bins ($k = 0, \dots, 8$; $\rho \leq \rho_{\text{cut}}$) and 8 high-frequency bins ($k = 9, \dots, 16$; $\rho > \rho_{\text{cut}}$). By contrast, Tiny ImageNet’s 64×64 inputs yield $\rho_{\max} = 32$ and $\rho_{\text{cut}} = 16$, with 17 low-frequency bins ($k = 0, \dots, 16$) and 16 high-frequency bins ($k = 17, \dots, 32$); this provides roughly twice the spectral resolution per band. Because each radial bin k averages over $|B_k|$ frequency cells (Eq. 7), the standard error of the per-bin power estimate scales as $\text{SE} \propto 1/\sqrt{|B_k|}$; at 32×32 , where bin cardinalities are 2–4× smaller than at 64×64 , individual bin estimates are correspondingly noisier. With only 9 low-frequency bins forming the denominator of $r_{H/L}$, small perturbations in low-band energy propagate more strongly into the ratio, inflating the variance of $\Delta_{H/L}$ across layers. Under these constraints, ResNet-50+AdamW’s anomalous $\Delta_{H/L} = +0.093$ ($p = <0.0001$, $n = 44$) is statistically detectable but small in absolute terms (only ~20% of the Tiny ImageNet SGD effect, +0.454), and the sign reversal relative to Tiny ImageNet (−0.016, n.s.) plausibly reflects how BN and LN redistribute energy differently when the entire representable spectrum is compressed into ≤ 17 bins. The companion SGD result (+0.032, $p = 0.879$) is consistent with zero; the robust positive gap seen at 64×64 and 224×224 does not survive the resolution collapse. These observations jointly motivate excluding CIFAR-100 from the primary BH family (§M.6): the 32×32 regime probes a fundamentally different spectral domain in which bin-level noise and axis compression preclude direct comparison with higher-resolution results.

M.4 CIFAR-100 Resolution Constraint

At 32×32 input resolution, CIFAR-100 Stage 4 feature maps fall below the 8×8 spatial threshold required for DCT-based spectral extraction (§3.1), restricting analysis to Stages 1–3 ($n = 44$ valid layer pairs per configuration). This constitutes a genuine limitation: deeper layers near the classification head may exhibit different spectral behavior, and the reduced sample size widens confidence intervals. Nevertheless, the surviving layers span sufficient depth to detect effects where they exist, as demonstrated by the significant AdamW result ($p < 0.0001$). The SGD non-significance is informative: it reveals that the BN–LN spectral gap observed at higher resolutions does not trivially transfer to compressed frequency domains. That the most extreme overfitting in our study (32.1 pp for AdamW+LN) does not prevent detection of a significant spectral effect under AdamW further dissociates spectral measurement from generalization quality.

1607 M.5 Architecture-Dependent Normalization Sensitivity

1608 Architectures differ in their sensitivity to normalization choice under SGD. ResNet-50+SGD pro-
1609 duces a robust positive $\Delta_{H/L}$ (+0.454, $p < 0.0001$), VMamba-S+SGD yields a negative $\Delta_{H/L}$
1610 (-0.708 , $p = 0.038$, BH-adjusted $p = 0.049$), MambaVision-T+SGD is inconclusive ($\Delta_{H/L} =$
1611 $+15.533$, $p = 0.203$, $n = 6$), and DeiT-S+SGD collapses entirely (§I.1). The optimizer-
1612 normalization spectral interaction is thus architecture-dependent rather than a simple CNN-specific
1613 effect: the CNN ResNet-50 shows robust positive bias, the pure SSM VMamba-S shows a significant
1614 negative estimate (BH-adjusted $p = 0.049$), and the hybrid MambaVision-T remains underpowered.
1615 Per-seed spectral measurements for all completed configurations are reported in Appendix D.1.

1616 M.6 Sensitivity of Borderline Corrections

1617 VMamba-S+SGD illustrates how false-discovery-rate control interacts with borderline significance:
1618 the raw $p = 0.038$ is significant at $\alpha = 0.05$, but BH adjustment at $m = 13$ yields $p_{\text{adj}} = 0.049$,
1619 just below the threshold. Because the adjusted p depends on the number of tests m , adding configu-
1620 rations alters the BH thresholds. Specifically, including the two CIFAR-100 ResNet-50 pairs would
1621 raise m from 13 to 15; at rank 10 of 15, the BH threshold becomes $\alpha_{\text{BH}} = (10/15) \times 0.05 = 0.033$,
1622 and VMamba-S+SGD’s raw $p = 0.038$ would exceed it, rendering the result non-significant. We ac-
1623 knowledge this fragility explicitly. The primary analysis uses TinyImageNet-only configurations as
1624 the BH family because CIFAR-100’s 32×32 resolution constitutes a qualitatively different spectral
1625 regime: the Nyquist limit restricts the maximum representable spatial frequency to 16 cycles, com-
1626 pressing the observable frequency axis and altering the BN–LN spectral dynamics (Appendix M.3).
1627 Pooling these resolution regimes into a single BH family would conflate distinct spectral phenomena.
1628 This instability does not affect the core claims: the cross-configuration sign test (5/5 non-positive un-
1629 der AdamW) excludes VMamba-S+SGD, and the SGD positive-bias conclusion rests on ResNet-50
1630 and EfficientNet-B0 (both $p_{\text{adj}} < 0.01$; Appendix C.1).

1631 N Additional Limitations

1632 N.1 LN+SGD Seed Sensitivity

1633 ResNet-50+SGD+LN trained across four seeds produces one collapsed run (seed 42, v3): validation
1634 accuracy falls to chance at epoch 52 (Appendix H). Three of four seeds (s0, s123, s7) converge suc-
1635 cessfully with per-seed $\Delta_{H/L}$ of +0.454, +0.498, and +0.335 respectively (Table 11); the pooled
1636 CI [+0.270, +0.498] does not cross zero ($p < 0.0001$, $n = 75$).

1637 **Mechanism hypothesis.** LN normalizes per-sample across all channels, eliminating the per-
1638 channel scale information that BN preserves through its channel-wise statistics. Under SGD’s uni-
1639 form learning rate, this loss of scale structure leaves training vulnerable to large gradient updates that
1640 push the network into degenerate basins. The stochastic nature of the failure (1 of 4 seeds collapses)
1641 is consistent with a basin-of-attraction boundary that some initializations approach more closely
1642 than others, rather than a systematic divergence. This is a recognized fragility of the LN+SGD com-
1643 bination on CNNs Ba et al. [2016]; BN+SGD with identical hyperparameters is stable across all
1644 seeds ($\sigma \approx 0.4$ pp; `res_resnet50_sgd_bn_{s0,s42,s123,s7}`).

1645 **Scope of impact.** The seed collapse does not affect BN-side measurements or any AdamW-based
1646 configuration. With three surviving seeds (s0, s123, s7), the LN pool achieves the target 3-seed
1647 design ($n=75$), and the sign and magnitude of $\Delta_{H/L}$ are consistent across all three seeds. The
1648 directional finding (positive $\Delta_{H/L}$ under SGD) is further supported by cross-architecture replica-
1649 tion on EfficientNet-B0+SGD ($\Delta_{H/L} = +0.253$, $p = 0.003$; `res_effnet_sgd_{bn,ln}_{s42}`;
1650 Appendix E.1) and cross-dataset replication on ImageNet-100 ($\Delta_{H/L} = +0.357$, $p < 0.0001$, two
1651 seeds; Appendix F.1).

1652 N.2 Surviving Seed Spectral Concordance

1653 The three surviving LN seeds yield per-seed $\Delta_{H/L}$ of +0.454 (s0), +0.498 (s123), and +0.335
1654 (s7); `res_resnet50_sgd_{bn,ln}_{s0,s123,s7}`. All three values are individually significant

1655 ($p < 0.0001$, $n = 25$ layers each) and share the same sign (positive); the BN-to-high-frequency
1656 shift is present in all surviving training runs. The pooled CI $[+0.270, +0.498]$ ($n = 75$) does not
1657 cross zero.

1658 Three independent seeds establish cross-run reproducibility: the concordance in sign and magnitude
1659 across s0, s123, and s7 confirms that the positive $\Delta_{H/L}$ under SGD is a stable property of the
1660 ResNet-50+SGD+BN configuration, not driven by any single training run. The BN leave-one-out
1661 analysis (Appendix K.4) further confirms that pooled $\Delta_{H/L}$ is stable under removal of any single
1662 BN seed (range $[+0.438, +0.471]$).

1663 **NeurIPS Paper Checklist**

1664 **1. Claims**

1665 Question: Do the main claims made in the abstract and introduction accurately reflect the
1666 paper’s contributions and scope?

1667 [Yes] The central claim (that optimizer regime is a consistently observed correlate of
1668 normalization-induced spectral bias direction) is supported by controlled experiments
1669 across six architectures, two primary optimizers, and corroborated by a cross-configuration
1670 directional consistency check (sign test, $m = 5$ distinct Tiny ImageNet configurations,
1671 $p = 0.031$; sensitivity analyses with $m = 4$ – 6 in main text and Appendix).

1672 **2. Limitations**

1673 Question: Does the paper discuss the limitations of the work performed by the authors?

1674 [Yes] Section 5 (Limitations paragraph) discusses: single-dataset scope (mitigated by
1675 IN100 scale-up), single-seed configurations, untested normalization variants (GroupNorm,
1676 RMSNorm), spatial resolution filtering, and the anti-conservative nature of layer-level boot-
1677 strap CIs.

1678 **3. Theory Assumptions and Proofs**

1679 Question: For each theoretical result, does the paper provide the full set of assumptions and
1680 a complete (correct) proof?

1681 [NA] This is an empirical diagnostic study with no theoretical claims or proofs.

1682 **4. Experiments**

1683 Question: Does the paper provide sufficient information needed to reproduce the experi-
1684 ments?

1685 [Yes] Section 3 details architectures, optimizers, hyperparameters, dataset, training dura-
1686 tion, and spectral extraction pipeline. Appendix provides additional statistical methods,
1687 cutoff sensitivity, and per-seed breakdowns.

1688 **5. Reproducibility**

1689 Question: Does the paper fully disclose all the information needed to reproduce the main
1690 experimental results of the paper to the extent that it affects the main claims and/or conclu-
1691 sions of the paper?

1692 [Yes] All hyperparameters, random seeds, dataset details (including the training data du-
1693 plication artifact in Appendix C), and spectral extraction parameters (batch size, evaluation
1694 mode, spatial resolution filter) are reported.

1695 **6. Open access to data, code, and instructions**

1696 Question: Does the paper provide open access to the data, code, and instructions needed to
1697 reproduce the main experimental results?

1698 [No] Code and trained checkpoints will be released upon acceptance. Tiny ImageNet and
1699 ImageNet are publicly available. Analysis code (spectral extraction, bootstrap CI compu-
1700 tation) will be released upon acceptance.

1701 **7. Experimental Setting/Details**

1702 Question: Does the paper specify all the training and test details?

1703 [Yes] Section 3.3 specifies batch size (128), epochs (200), warmup + cosine schedule,
 1704 optimizer hyperparameters, and seeds. Section 3.1 specifies the spectral extraction protocol
 1705 (10 batches of 64 images, evaluation mode, $\geq 8 \times 8$ spatial filter).

1706 **8. Experiment Statistical Significance**

1707 Question: Does the paper report error bars suitably and correctly defined or other appropri-
 1708 ate information about the statistical significance of the experiments?

1709 [Yes] Table 1 reports 95% BCa bootstrap CIs and p -values for all configurations. Sec-
 1710 tion 3.4 explicitly notes that layer-level CIs are anti-conservative due to within-model de-
 1711 pendence and provides a cross-configuration sign test as a supplementary directional con-
 1712 sistency check.

1713 **9. Experiments Compute Resources**

1714 Question: For each experiment, does the paper provide sufficient information on the com-
 1715 puter resources needed to reproduce the experiments?

1716 [Yes] All experiments run on NVIDIA RTX 5090 (32 GB) GPUs. Each training run takes
 1717 2–8 hours on a single GPU depending on architecture. Total compute is approximately 400
 1718 GPU-hours across all configurations and seeds.

1719 **10. Code Of Ethics**

1720 Question: Does the research conducted in the paper conform, in every respect, with the
 1721 NeurIPS Code of Ethics?

1722 [Yes]

1723 **11. Broader Impacts**

1724 Question: Does the paper discuss both potential positive societal impacts and negative
 1725 societal impacts of the work?

1726 [Yes] A Broader Impact section following the Conclusion discusses potential positive im-
 1727 pacts (informing architecture design, improving adversarial robustness research) and notes
 1728 the absence of foreseeable negative consequences from this diagnostic study.

1729 **12. Safeguards**

1730 Question: Does the paper describe safeguards that have been put in place for responsible
 1731 release of data or models?

1732 [NA] No new models or datasets are released; all training uses existing public datasets.

1733 **13. Licenses**

1734 Question: Are the creators or original owners of assets used in the paper properly credited?

1735 [Yes] All architectures, datasets, and optimizers are cited with their original publications.

1736 **14. New Assets**

1737 Question: Are new assets introduced in the paper well documented and is the documenta-
 1738 tion provided alongside the assets?

1739 [NA] No new datasets or pretrained models are released as assets.

1740 **15. Crowdsourcing and Research with Human Subjects**

1741 Question: For crowdsourcing experiments and research with human subjects, does the pa-
 1742 per include the full text of instructions given to participants?

1743 [NA] No human subjects or crowdsourcing involved.

1744 **16. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

1745

1746 Question: Does the paper describe potential risks incurred by study participants?

1747 [NA] No human subjects involved.